

University of Groningen

Mind the Hype

Van Dam, Nicholas T; van Vugt, Marieke K; Vago, David R; Schmalzl, Laura; Saron, Clifford D; Olendzki, Andrew; Meissner, Ted; Lazar, Sara W; Kerr, Catherine E; Gorchov, Jolie

Published in:
 Perspectives on Psychological Science

DOI:
[10.1177/1745691617709589](https://doi.org/10.1177/1745691617709589)

IMPORTANT NOTE: You are advised to consult the publisher's version (publisher's PDF) if you wish to cite from it. Please check the document version below.

Document Version
 Publisher's PDF, also known as Version of record

Publication date:
 2018

[Link to publication in University of Groningen/UMCG research database](#)

Citation for published version (APA):

Van Dam, N. T., van Vugt, M. K., Vago, D. R., Schmalzl, L., Saron, C. D., Olendzki, A., Meissner, T., Lazar, S. W., Kerr, C. E., Gorchov, J., Fox, K. C. R., Field, B. A., Britton, W. B., Brefczynski-Lewis, J. A., & Meyer, D. E. (2018). Mind the Hype: A Critical Evaluation and Prescriptive Agenda for Research on Mindfulness and Meditation. *Perspectives on Psychological Science*, 13(1), 36-61.
<https://doi.org/10.1177/1745691617709589>

Copyright

Other than for strictly personal use, it is not permitted to download or to forward/distribute the text or part of it without the consent of the author(s) and/or copyright holder(s), unless the work is under an open content license (like Creative Commons).

The publication may also be distributed here under the terms of Article 25fa of the Dutch Copyright Act, indicated by the "Taverne" license. More information can be found on the University of Groningen website: <https://www.rug.nl/library/open-access/self-archiving-pure/taverne-amendment>.

Take-down policy

If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

Downloaded from the University of Groningen/UMCG research database (Pure): <http://www.rug.nl/research/portal>. For technical reasons the number of authors shown on this cover page is limited to 10 maximum.

Mind the Hype: A Critical Evaluation and Prescriptive Agenda for Research on Mindfulness and Meditation

Perspectives on Psychological Science
2018, Vol. 13(1) 36–61
© The Author(s) 2017
Reprints and permissions:
sagepub.com/journalsPermissions.nav
DOI: 10.1177/1745691617709589
www.psychologicalscience.org/PPS



Nicholas T. Van Dam^{1,2}, Marieke K. van Vugt^{3†},
David R. Vago^{4†}, Laura Schmalzl^{5†}, Clifford D. Saron^{6†},
Andrew Olendzki^{7†}, Ted Meissner^{8†}, Sara W. Lazar^{9†},
Catherine E. Kerr^{10*}, Jolie Gorchov^{11†}, Kieran C. R. Fox^{12†},
Brent A. Field^{13†}, Willoughby B. Britton^{14†},
Julie A. Brefczynski-Lewis^{15†}, and David E. Meyer¹⁶

¹Melbourne School of Psychological Sciences, University of Melbourne; ²Department of Psychiatry, Icahn School of Medicine at Mount Sinai; ³Institute of Artificial Intelligence and Cognitive Engineering, University of Groningen; ⁴Osher Center for Integrative Medicine, Departments of Psychiatry and Physical Medicine & Rehabilitation, Vanderbilt University Medical Center; ⁵College of Science and Integrative Health, Southern California University of Health Sciences; ⁶Center for Mind and Brain, University of California, Davis; ⁷Integrated Dharma Institute, Amherst, Massachusetts; ⁸Center for Mindfulness, University of Massachusetts Medical School; ⁹Massachusetts General Hospital, Harvard Medical School; ¹⁰Department of Family Medicine, Warren Alpert Medical School at Brown University; ¹¹Silver School of Social Work, New York University; ¹²Department of Neurology and Neurological Sciences, Stanford University; ¹³Princeton Neuroscience Institute, Princeton University; ¹⁴Department of Psychiatry and Human Behavior, Warren Alpert Medical School at Brown University; ¹⁵Department of Physiology and Pharmacology, School of Medicine, West Virginia University; and ¹⁶Department of Psychology, University of Michigan

Abstract

During the past two decades, mindfulness meditation has gone from being a fringe topic of scientific investigation to being an occasional replacement for psychotherapy, tool of corporate well-being, widely implemented educational practice, and “key to building more resilient soldiers.” Yet the mindfulness movement and empirical evidence supporting it have not gone without criticism. Misinformation and poor methodology associated with past studies of mindfulness may lead public consumers to be harmed, misled, and disappointed. Addressing such concerns, the present article discusses the difficulties of *defining* mindfulness, delineates the proper scope of research into mindfulness practices, and explicates crucial *methodological* issues for interpreting results from investigations of mindfulness. For doing so, the authors draw on their diverse areas of expertise to review the present state of mindfulness research, comprehensively summarizing what we do and do not know, while providing a prescriptive agenda for contemplative science, with a particular focus on assessment, mindfulness training, possible adverse effects, and intersection with brain imaging. Our goals are to inform interested scientists, the news media, and the public, to minimize harm, curb poor research practices, and staunch the flow of misinformation about the benefits, costs, and future prospects of mindfulness meditation.

Keywords

mindfulness, meditation, psychotherapy, neuroimaging, contemplative science, adverse effects, media hype, misinformation

†These authors contributed equally to the present article and are listed in reverse alphabetical order.

*Cathy Kerr passed away, unexpectedly, during the revision of this article.

Corresponding Author:

Nicholas T. Van Dam, Melbourne School of Psychological Sciences, University of Melbourne, Parkville, VIC 3010, Australia
E-mail: nicholas.vandam@unimelb.edu.au

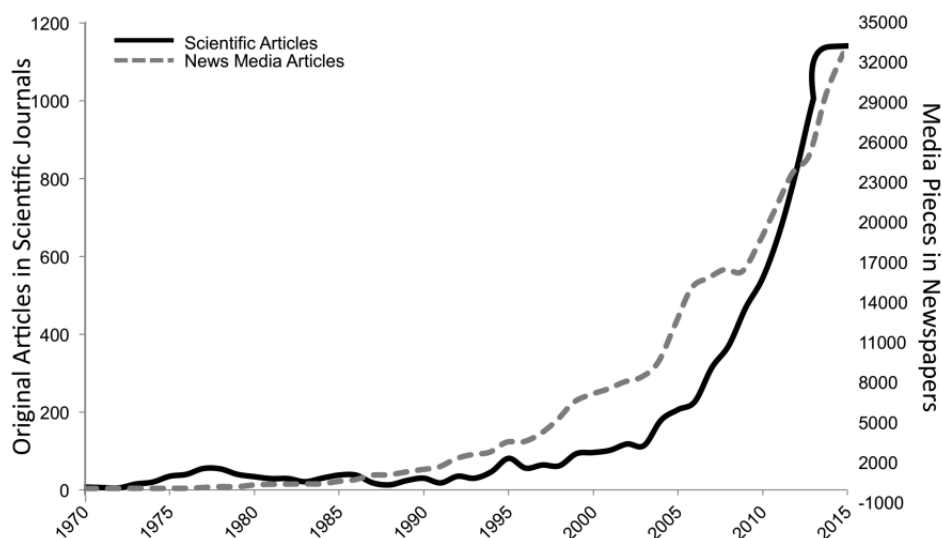


Fig. 1. Scientific and news media articles on mindfulness and/or meditation by year from 1970 to 2015. Empirical scientific articles (black line) with the term *mindfulness* or *meditation* in the abstract, title, or keywords, published between 1970 and 2015 were searched using Scopus. Media pieces (dashed gray line) with the term *mindfulness* or *meditation*, published in newspapers, using a similarity filter to minimize double-counting, published between 1970 and 2015 were searched using LexisNexis.

Mindfulness is an umbrella term used to characterize a large number of practices, processes, and characteristics, largely defined in relation to the capacities of attention, awareness, memory/retention, and acceptance/discernment. While the term has its historical footing in Buddhism (cf. Bodhi, 2011; Dreyfus, 2011; Dunne, 2011; Gethin, 2011; Kabat-Zinn, 2011), it has achieved wide-ranging popularity in psychology, psychiatry, medicine, neuroscience, and beyond, initially through its central role in mindfulness-based stress reduction (MBSR; Kabat-Zinn, 1990)—an intervention/training “package” introduced in the late 1970s as a complementary therapy for medically ailing individuals (Kabat-Zinn, 2011). The term *mindfulness* began to gain traction among scientists, clinicians, and scholars as the Mind and Life Institute emerged in 1987 and facilitated formal regular dialogues between the Dalai Lama and prominent scientists and clinicians, as well as regular summer research meetings, the latter starting in 2004 (Kabat-Zinn & Davidson, 2011). In the early 2000s, mindfulness saw an exponential growth trajectory that continues to this day (see Fig. 1). The term *mindfulness* has a plethora of meanings; a reflection of its incredible popularity alongside some preliminary support, considerable misinformation and misunderstanding, as well as a general lack of methodologically rigorous research.

Mindfulness has become an extremely influential practice for a sizeable subset of the general public, constituting part of Google’s business practices (Schaufenbuel, 2015), available as a standard psychotherapy via the National Health Service in the United Kingdom (see

Coyne, 2015b) and, most recently, part of standard education for approximately 6,000 school children in London (Rhodes, 2015). In addition, it has become a major area of study across subdisciplines of psychological science, including social/personality (Brown & Ryan, 2003), industrial/organizational (Dane, 2011), experimental (Jensen, Vangkilde, Frokjaer, & Hasselbalch, 2012), clinical (Dimidjian & Segal, 2015), cognitive (Tang, Hölzel, & Posner, 2015), health (Jain et al., 2007), educational (Britton, Lepp, et al., 2014), and many others. As such, it is critical that we take the term (along with any ambiguities) and the methodological rigor (or lack thereof) with which it has been studied very seriously.

Over the past two decades, writings on mindfulness and meditation practices have saturated the public news media and scientific literature (see Fig. 1). While this is not an isolated case, much popular media fail to accurately represent scientific examination of mindfulness (see, e.g., Goyal et al., 2014), making rather exaggerated claims about the potential benefits of mindfulness practices (Gibbs, 2016; Gunderson, 2016). There have even been some portrayals of mindfulness as an essentially universal panacea for various types of human deficiencies and ailments (see, e.g., Gunderson, 2016; Huffington, 2013).

As mindfulness has increasingly pervaded every aspect of contemporary society, so have misunderstandings about what it is, whom it helps, and how it affects the mind and brain. At a practical level, the misinformation and propagation of poor research methodology can potentially lead to people being harmed, cheated,

disappointed, and/or disaffected. At a philosophical level, misunderstandings of the work and its implications could limit the potential utility of a method that proposes unique links between first-person data and third-person observations (cf. Lutz & Thompson, 2003). Furthermore, research into a potentially promising arena may be halted for no reason other than that people have become tired of hearing about it (and therefore disinclined to pursue and/or fund it). While there have been many review articles written on mindfulness (e.g., Davidson & Kaszniak, 2015; Dimidjian & Segal, 2015; Farb, 2014; Tang et al., 2015), they cannot, by virtue of their limited scope (often focused on specific conditions or topics) and authorship (often limited to a short list of investigators, sometimes with clear conflicts of interest; see, e.g., Coyne, 2015b), offer a balanced, consensus perspective. Going beyond prior reviews, the present work provides exposition of the varying definitions of mindfulness, reviews the status of empirical assessment of mindfulness, reviews potential adverse events, considers implications for contemporary clinical practice, discusses specific issues that arise when doing neuroimaging with meditating samples, and elaborates on potential neural differences associated with meditation practices of varying durations.

Two main topics are considered herein: (a) the problem of *defining mindfulness* and thus delineating the appropriate scope of research on mindfulness practices and (b) *methodological issues* in mindfulness research. We provide (a) an overview of the current state in scientific knowledge, (b) a summary of consensus about what the currently available empirical findings do or do not conclusively show, and (c) a proposed prescriptive research agenda for making future scientific progress in understanding the consequences of mindfulness practices.

Our rationale for this expository approach stems from multiple major a priori considerations. We believe that much public confusion and media hype have stemmed from an undifferentiated use of the terms *mindfulness* and *meditation*. Each of these terms may refer to an ambiguously broad array of mental states and practices that are associated with a wide variety of secular and religious contexts (Davidson & Kaszniak, 2015; Goleman, 1988). Valid interpretation of empirical results from scientific research on such states and practices must take proper account of exactly what types of mindfulness and meditation are involved. With current use of umbrella terms, a 5-minute meditation exercise from a popular phone application might be treated the same as a 3-month meditation retreat (both labeled as meditation) and a self-report questionnaire might be equated with the characteristics of someone who has spent decades practicing a particular type of meditation (both labeled as mindfulness).

Furthermore, there is a general failure among the public to recognize that scientific consensus is a complex process requiring considerable time, effort, debate, and (most important) data. Throughout the scientific process, the predominant view among scholars can vacillate between being in support of, being agnostic to, and being against a given idea or theory (Shwed & Bearman, 2010). Eager journalists, academic press offices, and news media outlets—sometimes aided and abetted by researchers—have often overinterpreted initial tentative empirical results as if they were established facts. Moreover, statistically “significant” differences have repeatedly been equated with clinical and/or practical significance (cf. Rosnow & Rosenthal, 1989). These critical considerations need to be incorporated constructively in the future development of best practices for conducting mindfulness research, and for promoting accurate scientific communication with the general public (Britton, 2016).

The Problematic Meaning of “Mindfulness”

Despite how it is often portrayed by the media (e.g., Huffington, 2013) and some researchers (Brown & Ryan, 2003), there is neither one universally accepted technical definition of “mindfulness” nor any broad agreement about detailed aspects of the underlying concept to which it refers (Bodhi, 2011; Dreyfus, 2011; Dunne, 2011; Gethin, 2011). Frequently, “mindfulness” simply denotes a mental faculty for being consciously aware and taking account of currently prevailing situations (Kabat-Zinn, 1990; Langer, 1989). At other times, “mindfulness” may refer to formal practice of sitting on a cushion in a specific posture and attending (more or less successfully) to the breath or some other focal object. Considerable disagreement about definitions is not uncommon in the study of complex constructs (for discussion of intelligence, see, e.g., Neisser et al., 1996; for discussion of wisdom, see, e.g., Walsh, 2015) and mindfulness is no exception. Mindfulness is typically considered to be a mental faculty relating to attention, awareness, retention/memory, and/or discernment (cf. Davidson & Kaszniak, 2015); however, these multiple faculties are rarely represented in research practice (Goldberg et al., 2015; Manuel, Somohano, & Bowen, 2017). One of the most thoughtful and frequently invoked definitions states that mindfulness is moment-to-moment awareness, cultivated by paying attention in a specific way, in the present moment, as nonreactively, nonjudgmentally, and openheartedly as possible (Kabat-Zinn, 1990, 2011). However, this definition has been described as one of convenience regarding those constructs most readily comprehensible to Western audiences (Kabat-Zinn, 2011).

Table 1. Mindfulness Measures

Publication Date	Name	Context	Citation Count ^a	Factors
2001	Freiburg Mindfulness Inventory (FMI)	Buddhist theory	565	1. General
2003	Mindful Attention and Awareness Scale (MAAS)	Self-determination theory	5,054	1. Attentiveness and Awareness
2004	Kentucky Inventory of Mindfulness Skills (KIMS)	Dialectical behavior therapy	1,449	1. Observing 2. Describing 3. Awareness 4. Acceptance
2006	Five Facet Mindfulness Questionnaire (FFMQ)	CAMS-R, KIMS, FMI, SMQ, MAAS	2,660	1. Nonreactivity 2. Observing 3. Awareness 4. Describing 5. Nonjudging
2006	Toronto Mindfulness Scale (TMS)	Bishop et al. (2004)	648	1. Curiosity 2. Decentering
2007	Cognitive and Affective Mindfulness Scale, Revised (CAMS-R)	Buddhist theory and Kabat-Zinn (1990)	530	1. Attention 2. Present Focus 3. Awareness 4. Acceptance
2008	Philadelphia Mindfulness Scale (PHLMS)	Bishop et al. (2004)	411	1. Acceptance 2. Awareness
2008	Southampton Mindfulness Questionnaire (SMQ)	Kabat-Zinn (1990) and cognitive theory	297	1. General
2013	State Mindfulness Scale (SMS)	Buddhist theory	35	1. Body Mindfulness 2. Mind Mindfulness

^aGoogle Scholar, October 20, 2016.

Alternative semantic interpretations of “mindfulness”

Although concerted efforts have been made to provide consensus descriptions of mindfulness (Analayo, 2003; Bishop et al., 2004; Bodhi, 2011; Brown, Ryan, & Creswell, 2007; Grabovac, Lau, & Willett, 2011; Gunaratana, 2002; Hölzel et al., 2011; Malinowski, 2013; S. L. Shapiro, Carlson, Astin, & Freedman, 2006; Vago & Silbersweig, 2012), there continue to be considerable variations regarding the meaning of “mindfulness.” The resulting debates within and across complementary scholarly disciplines that encompass the investigation and practice of mindfulness and meditation more generally are diverse and complex (see *Contemporary Buddhism*, 2011, vol. 12, no. 1; *Psychological Inquiry*, 2007, vol. 18, no. 4). Given such considerations, one should not be especially surprised that some people have refrained from accepting Kabat-Zinn’s (1990) definition of “mindfulness,” or else have interpreted it in different, sometimes conflicting, ways. Kabat-Zinn (2011) himself has acknowledged that the term represents (to him) a much broader scope of concepts and practices than what his earlier (1990) definition might suggest.

Scientific implications of semantic ambiguity in the meaning of “mindfulness”

The ramifications of considerable semantic ambiguity in the meaning of mindfulness are multifarious. Any study that uses the term *mindfulness* must be scrutinized carefully, ascertaining exactly what type of “mindfulness” was involved, and what sorts of explicit instruction were actually given to participants for directing practice, if there was any practice involved. If the definition of mindfulness is based on self-report measures, one should be aware of the nuances of the various measures, how they relate to each other and/or conceptualizations of mindfulness (see Table 1; Bergomi, Tschacher, & Kupper, 2013; Sauer et al., 2013), as well as how different individuals might interpret the items on these measures (cf. Grossman & Van Dam, 2011). It should be further noted that self-reported mindfulness may not relate to the actual practice of mindfulness meditation (cf. Manuel et al., 2017). When formal meditation was used in a study, one ought to consider whether a specifically defined type of mindfulness or other meditation (cf. Lutz, Slagter, Dunne, & Davidson, 2008) was the target practice (see, e.g., Braun, 2013; McMahan, 2008).

In addition, while there is no single definition of mindfulness, it is important to examine whether the authors' specified definition is consistent with their study design.

Consequences of semantic ambiguity for empirical studies of “mindfulness”

Although most mindfulness training has been derived from the original MBSR model (Kabat-Zinn, 1990), the intensity (hours per day) and duration (total time commitment) of participants' formal practice have varied considerably across different versions of training (Davidson & Kaszniak, 2015; Tang et al., 2007; Zeidan et al., 2011). The particular methods for teaching and practicing “mindful” states have varied, too. However, published journal abstracts and media reports about obtained results often gloss over such crucial variations, leading to inappropriate comparisons between what might be fundamentally different states, experiences, skills, and practices.

Different definitions of skilled expertise. The definitions of “novice” and “expert” or “adept” (with respect to those with meditation experience) have varied considerably from study to study. Some investigators have considered novices to be individuals with some but not extensive prior formal meditation experience (e.g., up to a few hundred hours of practice; Kozasa et al., 2012; Lutz, Dunne, & Davidson, 2007). Others have applied a much stricter criterion, deeming novices only to be individuals with absolutely no prior meditation experience (e.g., Brewer et al., 2011). Further increasing this confusion, some approaches to investigating “mindfulness” (e.g., Hayes, Strosahl, & Wilson, 1999; Linehan, 1993) do not require any systematic training to become “skilled” in the practice, nor do they require participants to sustain a given experiential state (e.g., present-moment focus, or compassionate engagement) any longer than necessary to achieve a putative beneficial effect.

Consequences of semantic ambiguity for theoretical models of “mindfulness”

According to proposed theoretical models of mindfulness, there are clear mental processes and brain mechanisms that might facilitate insight and adaptive personal change, such as psychological distancing/reperceiving (S. L. Shapiro et al., 2006), decentering and inhibitory control (Vago & Silbersweig, 2012), nonconceptual discriminatory awareness (Brown et al., 2007), acceptance and reintegration (Hayes et al., 1999; Linehan, 1993),

or focused attention, decentering, and meta-awareness (Lutz, Jha, Dunne, & Saron, 2015; Meyer, 2009). Some of these processes and/or outcomes may be evident on a continuum, suggesting gradual growth with practice over time, whereas others may emerge significantly only in experienced practitioners (i.e., individuals who have engaged in formal sitting meditation or other contemplative practices such as hatha yoga, over a lengthy period of time; e.g., van Vugt & Slagter, 2014). Potential changes to various cognitive capacities as a result of mindfulness practice are not specific to clinical contexts; it also informs the limits, capacities, and nature of various cognitive functions and how those functions might be modified. However, the aforementioned complexity, confounding, and confusion that surrounds empirical research on “mindfulness” limits the potential of the method to inform broad questions and inform specific theories. The extent to which a specific model is supported or disconfirmed by particular sets of empirical data or systematic observations depends on the meaning of “mindfulness” that inspired data acquisition. For example, it is nearly impossible to test whether decentering has occurred if one has not obtained a measure of it. Support for a model will also depend on compliance with experimenter/clinician instructions (Davidson & Kaszniak, 2015). No one theoretical model (e.g., Garland, Farb, Goldin, & Fredrickson, 2015; Grabovac et al., 2011; Hölzel et al., 2011; S. L. Shapiro et al., 2006; Vago & Silbersweig, 2012) can possibly describe, explain, and predict all of the phenomena stemming from the panoply of facets that “mindfulness,” broadly construed, can have. Thus, it will be critical, going forward, to generate new integrative models and to track which data support which models.

Integrative assessment

Consensus about the semantic ambiguity of “mindfulness.” “Mindfulness” does not constitute a unitary construct, though it frequently includes aspects of paying attention in a specific, sustained, nonjudgmental way (Kabat-Zinn, 1990). Buddhist scholars suggest it often entails attention, awareness, memory/retention, and discernment (cf. Bodhi, 2011; Dreyfus, 2011; Dunne, 2011; Gethin, 2011). Self-report measures often highlight attention, awareness, and acceptance or nonjudgment (rather than discernment; see Table 1). The field, broadly defined, seems to agree that mindfulness entails attention and awareness with some important qualifiers about the nature of those faculties. It is also evident that mindfulness is part of some broader collection of goals and attitudes (Gethin, 2011; Kabat-Zinn, 2011). From a historical perspective, the attitudes qualifying attention and awareness are those accompanying some higher pursuit (e.g.,

Table 2. Nonexhaustive List of Defining Features for Characterization of Meditation Practice

Feature	Definition	Variation in Feature
Primary features		
Arousal	Extent of alertness, awakesness, etc.	Low, medium, high
Orientation (of attention)	Where attention is directed	Inward vs. outward vs. no orientation
Spatial “dynamic” (of attention)	The quality of attention in space	Fixed (e.g., on an object or location) vs. moving (e.g., as in the body scan)
Temporal “dynamic” (of attention)	The quality of attention in time	Constant/stable vs. rhythmic/sporadic
Object (of attention)	Attention can be fixed on none, one, two, or many objects	Specific (i.e., defined object[s]) vs. aspecific (i.e., no well-defined object[s]) vs. none (i.e., no object of attention)
Aperture (of attention)	How “sharply” the spotlight of attention is focused	Narrow vs. intermediate vs. diffuse
Effort	The extent to which one exerts energy to achieve other features	Low, medium, high
Secondary features		
Complementary activity	Physical activity to facilitate desired feature(s)	Walking, mantra recitation, dancing, rhythmic movement, etc.
Affective valence	Emotional tone of practice	Positive vs. neutral vs. negative
Emotional intention	A desired emotional state (to be cultivated)	Loving-kindness, compassion, forgiveness, generosity, etc.
Motivation/goal	The rationale/reason for the practice	Wellness, mitigation of illness, self-improvement, enlightenment
Proficiency required	Level of skill or expertise necessary	Low, medium, high
Posture	Physical orientation of body during practice	Horizontal (e.g., lying down) vs. intermediate (e.g., sitting) vs. vertical (e.g., standing)

enlightenment), including recognition/awareness, tranquility, concentration, equanimity, energy, joy, and discrimination (Gethin, 2011). Ultimately, degree of fidelity to historical definitions may not necessarily matter to definitions of mindfulness applied in modern practice (Dreyfus, 2011; Gethin, 2011), though historical definitions can provide important context and insight into the nature of mindfulness practice and its potential mechanisms (cf. Kabat-Zinn, 2011). Finally, the type of mindfulness putatively measured by contemporary cross-sectional research is not necessarily the same as what contemporary mindfulness training/meditation seeks to cultivate (see Manuel et al., 2017), which itself can differ from the mindfulness practiced by long-term meditators in various contemplative traditions relative to one another (Grossman & Van Dam, 2011).

Prescriptive research agenda: Transcending the prevalent ambiguity. Given current confusion surrounding “mindfulness,” we urge scientists, practitioners, instructors, and the public news media to move away from relying on the broad, umbrella rubric of “mindfulness” and toward more explicit, differentiated denotations of exactly what mental states, processes, and functions are being taught, practiced, and investigated. Toward this end, we

have provided a nonexhaustive list of defining features for characterization of contemplative and meditation practices (see Table 2). We have divided these features into primary (i.e., critical to most practices) and secondary (i.e., only critical to some practices). While this list is nonexhaustive, common use of this list of descriptors (or a comparable list) would permit the field to move beyond the many ambiguities of definition it is currently facing. Other examples of fundamental feature lists can be found in both scientific (e.g., Lutz et al., 2015) and contemplative (e.g., Analayo, 2003) literatures. For those studies using self-report measures, we encourage users to list the exact measure and to discuss the aspects of “mindfulness” that the utilized measure characterizes (see, e.g., Table 1). These suggestions address only terminology and do not necessarily provide ways to overcome the variation in the panoply of contextual factors surrounding mindfulness and/or meditation practice (e.g., type and training of instructor, regularity of meetings, group vs. individual practice, home practice type and amount). To resolve issues surrounding the implementation of mindfulness and/or other meditation-based training/intervention, we recommend development of something similar to a CONSORT checklist (Moher, Schulz, & Altman, 2001) that could be implemented across studies (see Table 3).

Table 3. Nonexhaustive List of Study Design Features for a Mindfulness-Based Intervention

Teacher information	Number/type of retreats attended? Experience in contemplative instruction (general and specific)? Formal contemplative training? Formal clinical qualifications? Blinded to experimental hypotheses?
Practice information	Setting(s)? Physical (e.g., hospital room, university lecture hall) Social (e.g., individual vs. group—if group, cohesion, size) Overall duration (e.g., 8 weeks, 12 weeks, 3 months)? Frequency of meetings? Average length of meetings? Types of formal practice (e.g., body scan, breath meditation, walking meditation)? Approximate total % of each type of practice? Types of informal practice? Logs maintained? Practice reviewed in session? Guided? Types of instructional materials used (e.g., mindfulness-based stress reduction workbook)
General information	Instructor adherence assessed? Control group used? Randomization/allocation method? Adverse events monitored?
Participant info	Inclusion/exclusion criteria? Prior meditation experience?
Conflicts of interest	Formal: funding agency Informal: Any possible financial benefit from results of study?

Methodological Issues in Mindfulness Meditation Research

Complementing our commentary about the problematic meanings of “mindfulness,” several major methodological issues in mindfulness meditation research should be considered as well. Such consideration is essential to achieve the present goals of providing a more balanced perspective on the pros and cons of practicing mindfulness, and on the weaknesses of currently available empirical findings about its efficacy. Specifically, we are concerned about four distinct but related types of issue: (a) insufficient construct validity in measures of mindfulness, (b) challenges to (clinical) intervention methodology, (c) potential adverse effects from practicing mindfulness, and (d) questionable interpretations of data from contemplative neuroscience concerning the mental processes and brain mechanisms underlying mindfulness.

Relation to the “replication crisis” in psychological science

Worries over scientific integrity and reproducibility of empirical findings have recently come to the fore of both psychological science and wider swaths of other basic and applied sciences, receiving considerable

attention in both the scientific literature (Button et al., 2013; Ioannidis, 2005, 2012; Miguel et al., 2014; Open Science Collaboration, 2012; Pashler & Wagenmakers, 2012) and public news media (Freedman, 2010; Johnson, 2014a, 2014b; Lehrer, 2010; Nyham, 2014). As part of these developments, debates regarding the efficacy and safety of treatment interventions have also embroiled the behavioral and neuropsychiatric sciences (Baker, McFall, & Shoham, 2008; Button et al., 2013; Fanelli, 2010; Ioannidis, 2005; Munafò, Stothart, & Flint, 2009; Simmons, Nelson, & Simonsohn, 2011; Yarkoni, Poldrack, Van Essen, & Wager, 2010). Although our present focus is on methodological issues to which mindfulness research is especially vulnerable, it is important to take account of this broader self-examination currently underway in the scientific community. Contemplative science (i.e., the scientific study of contemplative practices including, but not limited to, mindfulness meditation) is particularly vulnerable to “hype” of various sorts (i.e., tendencies to tout exaggerated positive and negative claims).

Insufficient construct validity in measuring mindfulness

One of the disclaimers on offer here concerns construct validity in measuring mindfulness. For obvious reasons,

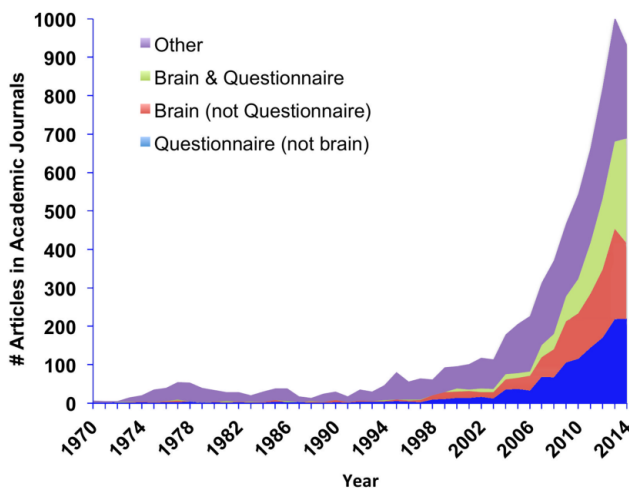


Fig. 2. Articles in academic journals by content type. Scopus search limited to articles in academic journals only, published between 1970 and 2014, keyword *mindfulness* or *meditation* for overall search; *Brain NOT Questionnaire* and *Questionnaire NOT Brain* as additional key terms.

this concern is crucial to our present objectives. Lacking reasonably validated mindfulness measures, one can neither properly determine how this mental faculty changes through instructions and guided practice, nor can one assess how increased mindfulness affects the cognitive capacities and/or symptoms of various mental and physical dysfunctions.

Difficulties in operationalizing and measuring mindfulness. Given the aforementioned absence of consensus regarding definitions of “mindfulness,” the operationalization and measurement of mindfulness are challenging endeavors. These difficulties have propagated to affect both (a) mindfulness practice and (b) assessments of mindfulness as a mental state or personality trait. Different researchers have implemented varying mindfulness training approaches across studies (e.g., Davidson, 2010), creating challenges for identifying common effects. We are especially concerned about attempts to measure mindfulness via self-report (see, e.g., Grossman & Van Dam, 2011) because, as Figure 2 indicates, a large fraction of recent research studies has used questionnaires for their primary assessment of mindfulness (consistent with a broader trend toward measuring psychological constructs via self-report; e.g., Baumeister, Vohs, & Funder, 2007).

Problematic aspects of self-report questionnaires. A major challenge to construct validity in psychological assessment is due to reluctance of the field to move beyond logical positivism, a philosophical position that suggests theories are direct derivations of that which can

be empirically observed (Green, 1992). Fueled by the prominence of behaviorism, which continues to play a prominent role in contemporary psychology (see, e.g., Plaud, 2001), the logical positivistic approach posits that a given measure is equivalent to the construct it purports to measure. In contrast, an alternative, nonjustificationist view suggests that a given measure is merely an approximation of a construct (Embretson, 1983; M. E. Strauss & Smith, 2009). It is important that philosophical views on construct validity can influence the ways that measures are designed and validated. One contemporary extension of logical positivism (which itself would reject the very idea of a construct) seems to be that nomothetic span (e.g., the extent to which a measure converges or diverges from other measures that are related or unrelated, respectively) is all that is needed for construct validity. In contrast to the positivistic view, construct representation (e.g., the psychological processes that give rise to responses on instruments that purport to measure the construct) is critical to construct validity (Embretson, 1983; M. E. Strauss & Smith, 2009).

Questionnaire-based scales that purport to measure mindfulness offer, at best, modest evidence of nomothetic span. Mindfulness does reliably correlate with other constructs such as emotional intelligence, self-compassion, psychological symptoms, thought suppression, emotion regulation, alexithymia, dissociation, and absent-mindedness (e.g., Baer, Smith, Hopkins, Kitemeyer, & Toney, 2006). However, these findings may actually be suggestive of a lack of differentiation from broad features of personality and temperament; meta-analysis of mindfulness measures suggests a strong negative relationship to neuroticism and negative affect (Giluk, 2009). As an alternative, it may suggest that at least some measures of mindfulness relate to general vulnerabilities or skills that are developed across interventions. In other words, these vulnerabilities and/or skills may not be specifically related to mindfulness, an idea supported by increases in mindfulness across both MBSR and an active control condition (Goldberg et al., 2015).

Additional psychometric concerns, largely relating to construct representation, about self-report mindfulness also exist. Notably, several of these scales exhibit different factor structures and response properties between meditators and nonmeditators (e.g., Christopher, Charoensuk, Gilbert, Neary, & Pearce, 2009; Van Dam, Earleywine, & Danoff-Burg, 2009), as well as before and after mindfulness training (e.g., Gu et al., 2016). These findings suggest lack of equivalence on a common underlying latent variable, as well as change in how the items are interpreted. One possible reason for this has to do with demand characteristics; one who has practiced mindfulness meditation may understand and value items differently than someone who has not practiced

(though see Baer, Samuel, & Lykins, 2011)—a potential conflation of desire to be “mindful” with actually being “mindful” (cf. Grossman, 2011). Of additional concern, mindfulness measures have not always favored the group one might expect to be more mindful; in one case, experienced meditators were less “mindful” than binge drinkers (Grossman & Van Dam, 2011; Leigh, Bowen, & Marlatt, 2005). Moreover, mindfulness questionnaires do not always correlate with mindfulness meditation practice (Manuel et al., 2017) and the underlying latent variable influencing item response on certain scales may be reflective of some general feature such as inattentiveness (Van Dam, Earleywine, & Borders, 2010).

Self-report-based measures of mindfulness may be particularly vulnerable to limitations of introspection because participants may not know exactly which aspects of mental states should be taken into account when making personal assessments. Moreover, making “on-line” judgments about degrees of mindfulness requires a special kind of multitasking (Meyer, 2009). In addition, social-desirability biases may be especially pronounced in self-reports about “mindfulness.” This is because participants/patients often learn to expect/value improved attention, equanimity, and so forth, while experimenters often fail to hide their hopes that participants will grow in their adeptness at these mental faculties (cf. Jensen et al., 2012).

Consensus about construct validity in measuring “mindfulness.” Some promise exists toward more accurate mindfulness measures via subjective report of behavioral indicators (e.g., breath counting; Frewen, Evans, Maraj, Dozois, & Partridge, 2007; Frewen, Lundberg, MacKinley, & Wrath, 2011; Levinson, Stoll, Kindy, Merry, & Davidson, 2014). Yet potential pitfalls exist even in these new measures (Ring, Brener, Knapp, & Mailloux, 2015). Although some self-report questionnaire measures of mindfulness seem to be effective in revealing particular mental and physical changes associated with practicing mindfulness (e.g., Baer, 2011), how closely these measures track exactly what is taught during practice remains unclear. While some investigators have implied that increased mindfulness improves the quality of participants’ introspections (Lutz et al., 2007; Mrazek, Smallwood, & Schooler, 2012; Zanesco, King, MacLean, & Saron, 2013), this claim has not been well established (cf. Fox et al., 2012; Levinson et al., 2014; Sze, Gyurak, Yuan, & Levenson, 2010; Whitmarsh, Barendregt, Schoffelen, & Jensen, 2014). Nor is it entirely obvious how one could veridically establish such a claim, for doing so would require accurate “third-person” evidence about the subjective contents of an introspector’s “first-person” consciousness (cf. Lutz, Lachaux, Martinerie, & Varela, 2002). It is ironic that were it shown that mindfulness practice

improves the quality of participants’ introspections, this might deepen other problems in mindfulness research. For example, if mindfulness-based enhancements of introspective accuracy are real, such enhancements could increase honest responding, thereby exacerbating between group confounds.

Perhaps because of such pitfalls in introspection, many studies have focused instead on neurobehavioral performance, attempting to assess mindfulness indirectly (e.g., Brewer et al., 2011; Ferrarelli et al., 2013; Jha, Krompinger, & Baime, 2007; Lao, Kissane, & Meadows, 2016; Lutz, Greischar, Perlman, & Davidson, 2009; Sahdra et al., 2011). However, these studies have inconsistent and sometimes contradictory empirical findings about the effects of mindfulness training on various basic cognitive and behavioral capacities (e.g., Jha et al., 2007; Lao et al., 2016). Some promising preliminary examples include studies that involved different types of mindfulness training leading to modest improvements in the efficiency of attention, orienting, and executive cognitive control after varying types of practice (Jha et al., 2007; Sahdra et al., 2011; Slagter et al., 2007; Tang et al., 2007; Van den Hurk, Gionmi, Gielen, Speckens, & Barendregt, 2010). Even when statistically significant, the magnitudes of observed cognitive effects stemming from mindfulness practices have been rather small (Chiesa, Calati, & Serretti, 2011; Sedlmeier et al., 2012).

Prescriptive research agenda: Measuring aspects of mindfulness. Given the cultural history and multitude of contextual variations in the term *mindfulness*, scientific research on the aggregate of mental states labeled by it would benefit from redirecting attempts to directly measure mindfulness toward measuring supporting mental faculties. The situation is similar to the psychological study of “intelligence.” Because of complexities, historical efforts to obtain a single unitary measure of general intelligence evolved to studying particular cognitive capacities, that, in combination, may make people functionally more or less intelligent (cf. Neisser et al., 1996).

Paralleling such evolution, we recommend that future research on mindfulness aim to produce a body of work for describing and explaining what biological, emotional, cognitive, behavioral, and social, as well as other such mental and physical functions change with mindfulness training. There are two broadly useful contexts in which to approach this problem. The first is to use a multimodal approach wherein first- and third-person (i.e., neurobiological and/or behavioral) assessments are used to mutually inform and identify one another (cf. Lutz et al., 2015; Lutz et al., 2002). This constitutes a more theory-driven approach to the problem of understanding mindfulness. A data-driven alternative might be comparable to how individuals in

affective neuroscience have used advanced algorithms to integrate physiological and neurobiological signals toward understanding emotional states (cf. Kragel & LaBar, 2014). A second context is to focus on the indirect impact of mindfulness practice, such as how meditation practice might lead to more effective therapists via assessing patient outcome (cf. Grepmaier et al., 2007) or how mindfulness might improve caregiver efficacy via assessment of significant others (cf. Singh et al., 2004). Another approach within this domain might be to examine how mindfulness practice can lead to changes in observable behaviors such as eating patterns or interpersonal exchanges (Papies, Pronk, Keesman, & Barsalou, 2015), the latter especially as reported by friends or partners of those undergoing mindfulness and/or meditation training (e.g., Birnie, Garland, & Carlson, 2010). In addition, researchers should situate future process models of mindfulness within extant rigorous theoretical frameworks for cognition and emotion whereby empirical predictions and falsifiable conceptual hypotheses can be tested (e.g., Meyer, 2009; Vago & Silbersweig, 2012). Frameworks based on computational modeling may be especially helpful for such purposes (e.g., Anderson et al., 2004; Meyer & Kieras, 1999).

Challenges for clinical intervention methodology

Numerous intervention studies have been conducted to assess whether, and by how much, practicing mindfulness may help alleviate various undesirable mental and physical conditions, including pain, stress, anxiety, depression, obesity, addiction, and others. Dimidjian and Segal (2015) estimate, using the NIH stage model for clinical science (Onken, Carroll, Shoham, Cuthbert, & Riddle, 2014), that only 30% of research using mindfulness-based interventions (MBIs) has moved beyond Stage 1 (intervention generation/refinement). The majority (20%) of research beyond Stage 1 has been conducted at Stage 2a (efficacy in research clinic: compared to wait-list control or treatment as usual), with a mere 9% (of the total) at Stage 2b (efficacy in research clinic: compared to active control). Moreover, only 1% of all research has been conducted outside research contexts, a woefully inadequate research base to inform whether MBIs are ready for use in regular clinical practice, as is the case in the United Kingdom (Coyne, 2015b, 2016). As a result, some have blatantly stated that “widespread use is premature” (Greenberg & Harris, 2012).

Haphazard variability across MBIs. Given the lack of consensus about what “mindfulness” means and how it should be operationalized, MBIs have varied greatly in

the diverse types of practice, methods of participant training, and duration of instructional courses associated with them. The “gold-standard model” of an MBI has been the 8-week mindfulness-based stress reduction (MBSR; Kabat-Zinn, 1990) course, involving 20 to 26 hours of formal meditation training during 8 weekly group classes (1.5–2.5 hours/class), one all-day (6 hours) class, and home practice (about 45 minutes/day, 6 days/week). Throughout the 8 weeks, formal MBSR training has included an eclectic set of specific mindfulness practices—focused attention on the breath, open monitoring of awareness in “body-scanning” (cf. Lutz et al., 2008), prosocial meditation (e.g., loving kindness and compassion), and gentle hatha yoga.

“Spin-off” MBIs vary in content and form depending on the participant populations for which they were adapted and the accompanying idiosyncratic objectives of individual investigators (cf. Shonin, Van Gordon, & Griffiths, 2013). For example, interventions such as mindfulness-based cognitive therapy (MBCT; Segal, Williams, & Teasdale, 2002) have incorporated aspects of cognitive behavioral therapy (CBT, widely considered the most researched and empirically based psychotherapy, focuses on the relationship between thoughts, emotions, and behaviors, most commonly with a focus on changing thought and behavioral patterns; Tolin, 2010). Notably, there are also a number of psychotherapies that draw on “mindful” principles, but are more commonly associated with traditional CBT (cf. Hofmann & Asmundson, 2008); these include acceptance and commitment therapy (Hayes et al., 1999) and dialectical behavior therapy (Linehan, 1993). We focus our discussions of MBIs on those interventions that utilize formal meditation techniques (namely, derivatives of MBSR), as they arguably differ in origin from those interventions more closely tied to cognitive and/or behavioral therapy (cf. Dimidjian & Segal, 2015; Hayes, 2002; Kabat-Zinn, 2011; Robins, 2002). Moreover, interventions that formally employ meditation practices differ in therapeutic delivery from those that do not formally employ such practices, though this distinction has become muddled as mindfulness and meditation have enjoyed greater mainstream popularity.

The duration of MBIs have been altered dramatically to conform with brief training regimens that may involve as few as four 20-minute sessions (e.g., Papies, Barsalou, & Custers, 2012; Zeidan et al., 2015). Some newer MBIs have even implemented web-based or mobile applications for treatment delivery (Cavanaugh et al., 2013; Dimidjian et al., 2014; Lim, Condon, & DeSteno, 2015). Given the variety of practices that fall under the umbrella of MBI, the adoption of mindfulness as a prescriptive clinical treatment has not entailed a consistent type of intervention. While there is considerable

variability in other practices of psychotherapy as well, specific classes of intervention (e.g., CBT) at least tend to have sufficient consistency with one another (in terms of content and format) to provide a basis for broad evaluation of their efficacy (cf. Tolin, 2010). In contrast, the varieties of interventions labeled as “mindful” are as varied as the definitions of the construct (differing in content, meeting type/frequency, instructions, homework, readings, instructor/therapist training and accessibility, etc.). Extreme caution must be exercised when considering mainstream implementation of minimally tested adaptations of more traditional MBIs (Dimidjian & Segal, 2015).

Misperceptions of therapeutic efficacy. Despite the preceding list of concerns, there is a common misperception in public and government domains that compelling clinical evidence exists for the broad and strong efficacy of mindfulness as a therapeutic intervention (e.g., Coyne, 2016; Freeman & Freeman, 2015). Results from some clinical studies conducted over the past 10 years have indicated that MBCT may be modestly helpful for some individuals with residual symptoms of depression (Eisendrath et al., 2008; Geschwind, Peeters, Huibers, van Os, & Wichers, 2012; van Aalderen et al., 2012). As a consequence of select results, published in high-profile journals, MBCT is now officially endorsed by the American Psychiatric Association for preventing relapse in remitted patients who have had three or more previous episodes of depression. Moreover, the U.K. National Institute for Health and Clinical Excellence now even recommends MBCT over other more conventional treatments (e.g., SSRIs) for preventing depressive relapse (Crane & Kuyken, 2012). Mitigating such endorsements, a recent meta-analysis found that MBSR did not generally benefit patients susceptible to relapses of depression (C. Strauss, Cavanagh, Oliver, & Pettman, 2014). Other meta-analysis have suggested general efficacy of MBIs for depressive and anxious symptoms (Hofmann, Sawyer, Witt, & Oh, 2010), though head-to-head comparisons of MBIs to other evidence-based practices have resulted in mixed findings, some suggesting comparable outcomes, others suggesting MBIs might be superior in certain conditions, and others suggesting CBT is superior in certain conditions (e.g., Arch et al., 2013; Goldin et al., 2016; Manicavasgar, Parker, & Perich, 2011). There is also mixed evidence comparing MBIs to interventions such as progressive muscle relaxation (e.g., Agee, Danoff-Burg, & Grant, 2009; Jain et al., 2007). Direct comparisons of MBIs to empirically established treatments are limited.

In a recent review and meta-analysis commissioned by the U.S. Agency for Healthcare Research and Quality, MBIs (compared to active controls) were found to have a mixture of only moderate, low, or no efficacy,

depending on the disorder being treated. Specifically, the efficacy of mindfulness was only moderate in reducing symptoms of anxiety, depression, and pain. Also efficacy was low in reducing stress and improving quality of life. There was no effect or insufficient evidence for attention, positive mood, substance abuse, eating habits, sleep, and weight control (Goyal et al., 2014). These and other limitations echoed those from a report issued just 7 years earlier (Ospina et al., 2007). The lack of improvement over these 7 years in the rigor of the methods used to validate MBIs is concerning; indeed if research does not extend beyond Stage 2A (comparison of MBI to wait-list control), it will be difficult, if not impossible, to ascertain whether MBIs are effective in the real world (cf. Dimidjian & Segal, 2015). On balance, much more research will be needed before we know for what mental and physical disorders, in which individuals, MBIs are definitively helpful.

Consensus about clinical intervention methodology. MBIs are sometimes misleadingly described as “comparable” to antidepressant medications (ADMs) (Goyal et al., 2014). Such comparability has been tentatively supported by results from studies examining MBIs versus ADMs for depressive relapse in recurrent depression (Kuyken et al., 2015; Segal et al., 2010). Notably, there are large individual differences in efficacy: MBIs may be beneficial for some people, but may be ineffective or contraindicated for others (Dobkin, Irving, & Amar, 2011). Special care is therefore needed when interpreting results from clinical studies employing MBIs, many of which have lacked “active” control conditions. Given the absence of scientific rigor in clinical mindfulness research (Davidson & Kaszniak, 2015; Goyal et al., 2014), evidence for use of MBIs in clinical contexts should be considered preliminary.

The official standards of practice for MBSR exclude suicidality and the presence of *any* psychiatric disorder (Santorelli, 2014). Case-by-case exceptions are permissible by these standards if, and only if, an individual participant is willing and able to simultaneously maintain adequate medical treatment for the exclusionary condition or if an instructor has sufficient clinical training to manage the case at hand (Santorelli, 2014). The American Psychiatric Association (D. H. Shapiro, 1982), the U.S. National Institutes of Health (NIH; National Center for Complementary and Integrative Health, 2016b), and leading researchers in the field (Dobkin et al., 2011; Greenberg & Harris, 2012; Lustyk, Chawla, Nolan, & Marlatt, 2009) have expressed concerns that meditation may be contraindicated under several circumstances. Numerous authors have recommended that schizophrenia spectrum disorders, bipolar disorder, post-traumatic stress disorder, depression, and risk factors for

psychosis (e.g., schizoid personality disorder) are contraindications to participation in an MBI that is not specifically tailored to one of these conditions (Didonna & Gonzalez, 2009; Dobkin et al., 2011; Germer, 2005; Kuijpers, van der Heijden, Tuinier, & Verhoeven, 2007; Lustyk et al., 2009; Manocha, 2000; Walsh & Roche, 1979; Yorston, 2001). The rationale for these contraindications is that without sufficient clinical monitoring, an intervention not designed to address these issues could lead to deterioration or worse. Such contraindications should be considered exclusionary criteria for regular clinical practice until substantially more evidence about the efficacy of various MBIs becomes available.

Prescriptive research agenda: Strengthening clinical intervention methods. Replication of earlier studies with appropriately randomized designs and proper active control groups will be absolutely crucial. In conducting this work, we recommend that researchers provide explicit detail of mindfulness measures (see, e.g., Table 1), primary outcome measures, mindfulness/meditation practices (see Table 2), and intervention protocol (see Table 3). While active control groups for MBIs can be difficult to implement for a variety of reasons (Davidson & Kaszniak, 2015), the problem is not insurmountable (see, e.g., MacCoon et al., 2012) and has been resolved by those conducting more traditional psychotherapy research (e.g., Agee et al., 2009; Arch et al., 2013; Goldin et al., 2016; Jain et al., 2007; Manicavasgar et al., 2011). In addition, researchers must be explicit about the exact hypothesis they are testing (noninferiority to an established treatment, superiority to an established treatment, etc.) and consider the various limitations that might accompany treatment designs (see, e.g., Coyne, 2015a).

Because of potential confirmation biases (Rosnow, 2002) and allegiance effects (Martin, Garske, & Davis, 2000), clinical research ideally would involve multidisciplinary teams of investigators. These teams should consist of not only clinicians, but also basic research scientists, scholars from within classical mindfulness traditions, and scientists/scholars skeptical about mindfulness's efficacy. An especially compelling research strategy could involve adversarial collaboration (see, e.g., Matzke et al., 2015). Moreover, future clinical studies should not rely merely on self-report and assessments by clinicians, but also incorporate biological and behavioral efficacy measures.

Harm, adverse effects, and fallout of meditation practices

Much of the public news media has touted mindfulness as a panacea for what ails human kind (e.g., Chan, 2013; Firestone, 2013), overlooking the very real

potential for several different types of harm. According to directors of the National Center for Complementary and Integrative Health (NCCIH) at the NIH, the biggest potentials for harm of complementary treatments (e.g., meditation) are “unjustified claims of benefit, possible adverse effects . . . and the possibility that vulnerable patients with serious diseases may be misled” (Briggs & Killen, 2013). Identifying “harm,” “side effects,” or “adverse effects” is complicated by issues related to definitions and measurement, which will be addressed in turn.

Coming to terms with meditation-related adverse effects. An adverse effect or event (AE) is any unwanted, harmful effect that results from but is not the stated goal of a given treatment. A side effect is any unexpected effect that is secondary to the intended effect of the treatment (M. Linden, 2013). An event can also be categorized a “side effect” if it is not described in the “product labeling,” “package insert,” “marketing or advertising” (NIA, 2011; Office for Human Research Protections, 2007)—descriptions that are often lacking for meditation practices (and behavioral interventions more generally, despite a comparable incidence of AEs to pharmacological treatments; Crawford et al., 2016; M. Linden, 2013; Mohr, 1995; Moos, 2005, 2012). Whether the result of correct or incorrect treatment, a treatment-emergent reaction may include the appearance of novel symptoms that did not exist before treatment, or the exacerbation or reemergence of a preexisting condition. Treatment nonresponse or deterioration of (target) illness may or may not be caused by the treatment (M. Linden, 2013) but requires both reporting and action.

Meditation-related experiences that were serious or distressing enough to warrant additional treatment or medical attention have been reported in more than 20 published case reports or observational studies. These reports document instances of meditation-related or “meditation-induced” (i.e., occurring in close temporal proximity to meditation and causally attributed to meditation by the practitioner, instructor, or both) psychosis, mania, depersonalization, anxiety, panic, traumatic-memory reexperiencing, and other forms of clinical deterioration (Boorstein, 1996; Carrington, 1977; Castillo, 1990; Chan-Ob & Boonyanaruthee, 1999; Disayavanish & Disayavanish, 1984; Epstein & Lieff, 1981; Heide & Borkovec, 1983; Kerr, Josyula, & Littenberg, 2011; Kornfield, 1979; Kuijpers et al., 2007; Kutz et al., 1985; Lomas, Cartwright, Edginton, & Ridge, 2015; Miller, 1993; Nakaya & Ohmori, 2010; Sethi, 2003; D. H. Shapiro, 1992; Shonin, Van Gordon, & Griffiths, 2014b, 2014c; VanderKooi, 1997; Van Nuys, 1973; Walsh & Roche, 1979; Yorston, 2001). Many of the aforementioned were case studies, case series, or observational studies, often without a control group. Only one was

prospective (D. H. Shapiro, 1992). Detailed clinical histories were available for some of the subjects, but not all, which makes the question of preexisting conditions difficult to evaluate. While qualitative reports and case studies are an appropriate and necessary first step in identifying potential AEs (Dimidjian & Hollon, 2010), the need for AE assessments within more rigorous designs such as randomized controlled trials (RCTs) would provide more conclusive information.

Issues in the measurement of adverse effects. Since safety reporting is required for federally funded clinical trials, one might expect that the many NIH-funded mindfulness or meditation trials would be a rich source of information about potential AEs with causality assessment inherent in an RCT design. However, most current methods for assessing AEs in meditation-related research are insufficient to produce an accurate estimate. Despite CONSORT requirements (Moher et al., 2001), and compared to 100% of pharmacology trials (Vaughan, Goldstein, Alikakos, Cohen, & Serby, 2014), less than 25% of meditation trials actively assess AEs (Goyal et al., 2014; Jonsson, Alaie, Parling, & Arnberg, 2014), relying instead on spontaneous reporting, which may underestimate AE frequency by more than 20-fold (Bent, Padula, & Avins, 2006), and results in widely varying AE rates, even for similar trials (Kuyken et al., 2015; Kuyken et al., 2016; J. M. Williams et al., 2014). Different AE assessment methods (Vaughan et al., 2014) or specifically the lack of systematic AE assessment in meditation trials has led to the hasty and erroneous conclusion not only that meditation is free of AEs (L. Turner et al., 2011), but also that meditation interventions can act as a replacement to medication for mental illnesses such as depression and bipolar disorder (Annels, Kho, & Bridge, 2016; Strawn et al., 2016; Walton, 2014) with slogans such as “meditate not medicate” (Annels et al., 2016). Furthermore, meditation-related AEs are discussed in many traditional (largely Buddhist) meditation guides (Buddhaghosa, 1991; Sayadaw, 1965; B. Wallace, 2011). Despite the assumption of “wide acceptance of minimal, if any, AEs associated with meditation” (L. Turner et al., 2011), this assumption is largely based on a lack of research rather than substantive evidence.

Other potential risks of mindfulness meditation. The benefits and the safety of meditation are likely exaggerated beyond available evidence in a manner that increases “the possibility that vulnerable patients with serious diseases may be misled” (Briggs & Killen, 2013). In the face of such exaggerated claims, patients may be diverted from pursuing other, more traditional activities (e.g., regular aerobic exercising) that typically yield physical and mental benefits (Cotman, Berchtold, & Christie, 2007; Penedo & Dahn, 2005) or standard treatments (e.g.,

psychotherapy, pharmacotherapy) that are better suited to dealing with particular psychiatric conditions. For example, in a recent meta-analysis of MBIs, C. Strauss et al. (2014) concluded, “given the paucity of evidence in their favour, we would caution against offering MBIs as a first line intervention for people experiencing a primary anxiety disorder . . . findings from the current meta-analysis would suggest great caution if offering MBIs to this population as a first line intervention instead of a well-established therapy.” In economics, as well as recent discussions of psychotherapy, this effect has been labeled an “opportunity cost” (i.e., time and money invested in a treatment approach that has little to no therapeutic benefit relative to the potential time/money that could have been invested in a treatment more likely to yield improvement; cf. Lilienfeld, Lynn, & Lohr, 2003). Given that relief from anxiety is probably one of most widely promoted benefits of mindfulness (see, e.g., Hofmann et al., 2010), opportunity cost may be a widespread “side effect” of MBI hype.

Consensus about harm, adverse effects, and contraindications. To date, “official” clinical guidelines about the state of meditation-related risks are in their infancy and only a handful of organizations and regulatory agencies have issued any statements. The American Psychiatric Association (APA) first showed concern about meditation-related AEs in 1977 and commissioned a report on the topic with treatment guidelines (D. H. Shapiro, 1982). The APA also included descriptions of meditation-induced depersonalization and other clinically relevant problems in both the 4th and 5th editions of their *Diagnostic and Statistical Manual of Mental Disorders* (APA, 1994, 2013). The NIH states that “meditation could cause or worsen certain psychiatric problems” but does not provide any practice guidelines beyond a boilerplate disclaimer to “check with your doctor” before trying meditation (NCCIH, 2016b).

Since neither meditation writ large nor meditation-based interventions are overseen by any regulatory agencies, most of the clinical guidelines and recommendations regarding risk and safety have been issued by the “Centers for Mindfulness,” creators of interventions, as well as various experts in the field. Many meditation researchers and clinicians have offered reviews of meditation-related risks, AEs, or contraindications with recommendations for clinical guidelines (Dobkin et al., 2011; Fenwick, 1983; Greenberg & Harris, 2012; Hanley, Abell, Osborn, Roehrig, & Canto, 2016; Lustyk et al., 2009; D. H. Shapiro, 1982; Shonin, Van Gordon, & Griffiths, 2014a). The MBCT Implementation Resources (Kuyken, Crane, & Williams, 2012) is one of the first documents to list potential “risks to participants,” including increased likelihood of suicidality,

depression, negative emotions, and flashbacks during meditation for individuals with trauma histories. At present, management strategies for potential risks have been largely limited to exclusion and informed consent. Both the University of Massachusetts Center for Mindfulness and the Oxford Mindfulness Centre have published recommended exclusion criteria for standard MBSR and MBCT, both excluding current suicidality and/or any current psychiatric disorder (Kuyken, Crane, & Williams, 2012; Santorelli, 2014). In addition, many centers attempt to make clear that mindfulness is not intended to replace standard psychiatric care.

Prescriptive research agenda: Transcending adverse effects. The current guidelines, while preliminary, represent substantial progress in assessing and promoting safety of meditation-based interventions. On the measurement front, there have been signs of progress. A few MBI researchers have started to actively monitor AEs either through questionnaires or through clinician interviews (Kuyken et al., 2015; Kuyken et al., 2016; J. M. Williams et al., 2014). While these are typically limited to serious AEs (life-threatening or fatal events) or “deterioration” on preexisting clinical outcomes that require clinical attention, such as increased depression or suicidality, this is a considerable improvement from passive monitoring.

In addition, a recent qualitative study of 60 Buddhist meditators and meditation teachers (cf. Lindahl et al., 2017) also sought to improve knowledge of meditation-related experiences that are underreported, unexpected, “adverse,” or associated with significant levels of distress and functional impairment. While qualitative and retrospective, this study applied 11 of the 13 causality criteria (as outlined by the World Health Organization [WHO], Federal Drug Administration, and NIH; Agbabiaka, Savovic, & Ernst, 2008; NIH, 2016; WHO, 2016), including interviews with meditation teachers (expert judgment). The study produced 60 categories of meditation-related experiences and 26 categories of “influencing factors” that may impact the duration, associated distress, and impairment of the experience. While the first study of its kind, it sets a foundation for testable hypotheses in future research. In addition, the 60 categories of meditation-related experiences are being converted into a measurement tool that can be used for systematic assessment across multiple studies and conditions. The codebook was inserted as an interview-based assessment into a recently completed clinical dismantling trial of MBCT (NCT no. 01831362) that can assess whether similar experiences occur in MBIs, as well as address the question of biological gradient (i.e., whether more exposure results in greater effects; Hill, 1965).

The large and growing body of empirical data on the psychological and neurobiological effects of meditation and related practices also represent a step forward to

identifying potential mechanisms by which meditation-related effects, as well as AEs might occur. Knowledge of mechanism may help identify who is at risk. For example, there is some evidence that hyperconnectivity of the prefrontal cortex and limbic regions may result in affective and autonomic blunting which is characteristic of dissociation (Ketay, Hamilton, Haas, & Simeon, 2014; Sierra et al., 2002). Similarly, increased activity in the inferior parietal cortex, a common outcome of mindfulness training (Brefczynski-Lewis, Lutz, Schaefer, Levinson, & Davidson, 2007; Farb et al., 2007; Goldin & Gross, 2010; Hasenkamp, Wilson-Mendenhall, Duncan, & Barsalou, 2012), might relate to depersonalization (disembodiment, loss of agency and self-other/self-world boundaries; Bunning & Blanke, 2005). Others have created neurobiological models for specific meditation-related experiences, such as visual hallucinations, (Lindahl, Kaplan, Winget, & Britton, 2014), sleep-related changes (insomnia; Britton, Lindahl, et al., 2014), changes in sense of self (Dor-Ziderman, Berkovich-Ohana, Glicksohn, & Goldstein, 2013), and altered perceptions of space and time (Berkovich-Ohana, Dor-Ziderman, Glicksohn, & Goldstein, 2013).

Research on AEs of treatments that share mechanisms with meditation should also be considered. For example, treatments that restrict environmental stimulation or narrative processing through internal sensory focus, such as qigong (APA, 2000; Shan, 2000), autogenic training (W. Linden, 1990), and relaxation (Edinger & Jacobsen, 1982), can precipitate similar AEs, such as autonomic hyperarousal, perceptual disturbances (Lindahl et al., 2014), traumatic memory reexperiencing (Brewin, 2015; Brewin, Gregory, Lipton, & Burgess, 2010; Miller, 1993), and psychosis (APA, 2000; Shan, 2000). Relaxation-induced panic or anxiety is perhaps one of the most well-documented phenomena with clear relevance to meditation (Adler, Craske, & Barlow, 1987; Cohen, Barlow, & Blanchard, 1985; Heide & Borkovec, 1983).

Challenges for investigating mindfulness through contemplative neuroscience

As part of the burgeoning trend in research on mindfulness and meditation more generally (Fig. 1), investigators have increasingly used methods from cognitive neuroscience, especially functional magnetic resonance imaging (fMRI). These methods yield visual depictions of participants' relative, regionally localized, brain activation during various types of cognitive task performance as well as the integrated functional neural networks of mental processing (including the default mode network; cf. Power et al., 2011). The investigation of mindfulness through such methods has also come

to be known as contemplative neuroscience (e.g., Davidson & Lutz, 2008).

Limitations in depictions of brain activity based on neuroimaging. Representative pictures from fMRI and other neuroimaging methods do not clearly convey the complex—often fraught—chain of biological and computational steps that lead to inferences about changes in brain structure and function. They also neglect to highlight the fact that such inferences are frequently derived from averages obtained across groups of participants. Thus, when also accompanied by numerous other difficult experimental, statistical, and inferential challenges prevalent in psychological research, contemplative neuroscience has often led to overly simplistic interpretations of nuanced neurocognitive and affective phenomena. For example, psychologist Rick Hanson, in what is presumably an effort to explain how meditation has been shown to influence emotion regulation, correlated with alterations in amygdala activity (e.g., Goldin & Gross, 2010), has stated, “In terms of amygdala activity, people seem to belong to one of three groups . . . the ones with a joyful amygdala—are more focused on promoting the good than on preventing the bad” (Hanson, 2013, pp. 43–44). As a result of such oversimplifications, meditative benefits may be exaggerated and undue societal urgency to undertake mindfulness practices may be encouraged (e.g., Farias & Wikholm, 2015).

Problematic aspects of group-level neuroimaging analyses. Furthermore, results from neuroimaging during mindfulness practices and other types of meditation may be subject to unique confounds. Despite variability in different types of practice and meditative experiences, it is not uncommon for neuroimaging data obtained from diverse practitioners to be pooled in aggregated analyses (e.g., Ferrarelli et al., 2013; Luders et al., 2012; Luders, Kurth, Toga, Narr, & Gaser, 2013; Sperduti, Martinelli, & Piolino, 2012). Also complicating theoretical interpretation of their results and further adding to confounds associated with systematic individual differences, many neuroimaging studies have used cross-sectional designs, precluding possible inferences about underlying cause-and-effect relationships (cf. Tang et al., 2015).

Ancillary physical artifacts in neuroimaging data. Certain methodological confounds that plague neuroimaging studies in general, are of particular concern in studies of individuals who meditate. Physical artifacts involving head movements and cardiorespiratory effects are especially notable (Holmes, Solomon, Cappa, & Greenberg, 1983; Lutz et al., 2009; Reuter et al., 2015; Van Dijk, Sabuncu, & Buckner, 2012; R. K. Wallace, 1970; R. K. Wallace, Benson, & Wilson, 1971; cf. Lazar et al., 2000; Zeidan et al., 2011). If nonmeditators are more restless or

breathe more rapidly than experienced meditators during MRI sessions, there could be spurious group differences in some neuroimaging measurements (e.g., with respect to meditators, seemingly more brain gray matter and brain activation in particular neuroanatomical regions; cf. Greene, Black, & Schlaggar, 2016). Systematic individual differences in cardiorespiratory activity between nonmeditators and meditators are especially worrisome because of the so-called “vein-drain problem” (R. Turner, 2002). It prevails especially in typical regions of differential brain activation. Enlarged blood vessels may lead to measurement artifacts (e.g., Boubela et al., 2015), which can be particularly pronounced in brain regions commonly identified as important for cognition and emotion (e.g., insular and anterior cingulate cortices).

Partially mitigating these concerns, meta-analyses of both structural and functional neuroimaging data have revealed differences in brain regions that tend to be consistent with the specific meditation practices under study (e.g., changes in brain regions associated with bodily awareness of mindfulness practitioners—for example, the insula and somatosensory cortices—and widespread recruitment of brain regions associated with vision during meditative visualization). Such findings, when supported by results from meta-analyses of multiple studies, are less likely to have stemmed merely from artifacts (Fox et al., 2016; Fox et al., 2014).

Practical versus statistical significance of neuroimaging data. Statistical and theoretical approaches to calculating and interpreting effect sizes and associated confidence intervals have been well developed in behavioral and psychological research (Cumming, 2014). Yet calculating valid estimates of effect sizes in neuroimaging data is extremely difficult (Fox et al., 2016; Fox et al., 2014; Friston, 2012; Hupé, 2015). Consequently, the practical significance and clinical importance (e.g., diagnostic and/or therapeutic utility) of observed changes in brain structure and neural activity associated with practicing mindfulness is still elusive (cf. Castellanos, Di Martino, Craddock, Mehta, & Milham, 2013). Moreover, despite some agreement among investigators that mindfulness and other types of meditation affect the brain, we still do not know how the effects compare to other cognitive training methods regarding practical significance.

Consensus about findings from contemplative neuroscience. Despite the many serious limitations mentioned previously, studies in contemplative neuroscience do allow some preliminary conclusions. Meta-analyses of neuroimaging data suggest modest changes in brain structure as a result of practicing mindfulness (Fox et al., 2014). Some concomitant modest changes also have been observed in neural function (e.g., Fox et al., 2016; Sperduti et al., 2012; Tomasino, Fregona, Skrap, & Fabbro, 2013;

for a broad review, see Tang et al., 2015). Caution must be exerted in interpreting these findings; similar changes have been observed following other forms of mental and physical skill acquisition, such as learning to play musical instruments and learning to reason, suggesting that they may not be unique to mindfulness or other popular types of meditation practice (cf. Draganski & May, 2008; Hyde et al., 2009; Mackey, Miller Singley, & Bunge, 2013; Münte, Altenmüller, & Jäncke, 2002).

Prescriptive research agenda: Truth in advertising by contemplative neuroscience. Rather than contributing to further media hype, researchers in contemplative neuroscience must endeavor to communicate more accurately with other scientists, journalists, and the public not only about the potential benefits of mindfulness practices for mental processes and brain mechanisms, but also about the limitations of neuroimaging methods and data collected through them. We encourage contemplative neuroscientists to follow best practices in neuroimaging methods generally (cf. Nichols et al., 2017), but also to consider and accommodate unique issues that may arise while collecting brain data from meditating populations. These unique issues (e.g., different respiration rates, different cardiac activity, dramatically different demographic and life-style characteristics) may warrant unique data collection methods (e.g., cardiac-gated image acquisition) and/or analytic methods (e.g., removal of activity because of respiratory artifact), as well as very detailed demographic information. Particular attention should be paid to methodologically and/or statistically controlling potential contributions from potentially confounding variables (e.g., participant motivation, placebo effects, cardiorespiratory factors, head motion, history of psychopathology) that may underlie apparent group differences. This will be especially necessary where mindfulness studies compare results from long-term practitioners versus meditation-naïve participants. In contexts of comparing meditation experience, either between groups, or within, some common metric should be used (cf. Hasenkamp & Barsalou, 2012). Researchers should stress specifically that individuals who already have meditated over many years, or who—though not yet experts—are personally attracted to meditation, may have characteristics that differentiate them from the general population even before experimentation (Mascaro, Rilling, Negi, & Raison, 2013). Prominent mention about the limitations and fraught nuances of statistical neuroimaging analyses should not be neglected either. No amount of sophisticated statistical prowess can correct results from faulty or confounded methods, a fact of which researchers, scientists, and the public should regularly be reminded.

And, ultimately, the popular news media—inspired by honest, forthright, thorough cooperation with

contemplative neuroscientists—must persuade the general public together with government funding agencies that multiple large, longitudinal RCTs that consider participant preferences concerning mindfulness practices are required and should be funded. We need such trials to definitively determine the full benefits and costs of practicing mindfulness. Without future RCTs, prevalent widespread uncertainties surrounding past results from haphazard studies of mindfulness involving relatively small sample sizes (e.g., Button et al., 2013) and considerable variation in how neuroimaging methodologies have been implemented (Simmons et al., 2011) make it difficult to know the neural effects of mindfulness.

Conclusion

Contemplative psychological scientists and neuroscientists, along with other researchers who study mental processes and brain mechanisms underlying the practice of mindfulness and related types of meditation, have a considerable amount of work to make meaningful progress. Much work should go toward improving the rigor of methods used, along with the accuracy of news media publicity and eliminating public misunderstandings caused by past undue “mindfulness hype.” These efforts have to take place on several related fronts.

First, as mentioned before, the various possible meanings of “mindfulness” have to be clarified. To deal with prevailing inherent semantic ambiguities, researchers should adopt more nuanced, precisely focused, terminology for referring to the various distinct mental and physical states as well as overt behaviors often associated with mentions of “mindfulness” (see Table 2). Insofar as future research involves self-report questionnaires about mindfulness, new ones that incorporate specific terminology (see, e.g., Table 2) ought to be developed. Theoretical models formulated to account for data need also consider these new key terms.

Second, future studies of mindfulness should conform to lessons being learned from the ongoing “replication crisis” in psychological science and other related scientific disciplines. For example, preregistered experiments and open-science replications of mindfulness are desirable. Additional discipline is especially needed in light of recent growing troublesome meta-analytic evidence that—like some other “glitzy” popular topics of psychological and neural investigations—past mindfulness research has succumbed to these questionable practices (Coronado-Montoya et al., 2016).

Third, future clinical applications involving MBIs must seek to attain more uniformity and better control (see Table 3), especially where definitive answers have

yet to be found. It is critical that those who conduct clinical research provide warnings regarding the extent to which their research findings generalize to clinical practice. Also researchers and clinicians have to be put on guard, educated about, and encouraged to address the potential AEs stemming from mindfulness practices. Research on the nature and scope of potential AEs should receive considerable further attention and government funding, because of the public's rapidly increasing involvement in practicing mindfulness.

Fourth, as they continue to emerge through technological advances in neuroimaging methods, new findings from contemplative neuroscience about the mental processes and brain mechanisms of mindfulness practices must be reported with all due modesty. Their importation into protocols for future clinical practice must await proper vetting of the potential practical significance that may accompany them. This vetting process will have to deal diligently with the many aforementioned challenges that still remain to be surmounted by the contemplative neuroscience community.

Only with such diligent multipronged future endeavors may we hope to surmount the prior misunderstandings and past harms caused by pervasive mindfulness hype that has accompanied the contemplative science movement.

Acknowledgments

We dedicate this article to our dear friend and colleague, Cathy Kerr, who passed away unexpectedly during revision of this work. Cathy was among the key driving forces that led to this particular group forming and to our formal meeting in Amherst, Massachusetts, in July 2014. Cathy touched so many lives and had a profound influence on the variety of ways that many of us approach mindfulness and meditation research. She will be profoundly missed.

This article grew out of a series of conferences and workshops generously funded by the Mind and Life Institute. However, the views expressed here are those of the authors alone, collectively, and do not necessarily represent the views or policies of the Mind and Life Institute nor any other organizations with which the authors are affiliated. We would further like to add that while all authors contributed to the article, it should be read as a majority consensus; not all authors strictly ascribe to all statements contained herein.

Declaration of Conflicting Interests

The authors declared that they had no conflicts of interest with respect to their authorship or the publication of this article.

References

- Adler, C., Craske, M., & Barlow, D. (1987). Relaxation-induced panic: When resting isn't peaceful. *Integrative Psychiatry*, 9, 94–112.

- Agabiakia, T. B., Savovic, J., & Ernst, E. (2008). Methods for causality assessment of adverse drug reactions: A systematic review. *Drug Safety*, 31, 21–37.
- Agee, J. D., Danoff-Burg, S., & Grant, C. A. (2009). Comparing brief stress management courses in a community sample: Mindfulness skills and progressive muscle relaxation. *Explore: The Journal of Science and Healing*, 5, 104–109.
- American Psychiatric Association. (1994). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: Author.
- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed., text rev.). Washington, DC: Author.
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Washington, DC: Author.
- Analayo, B. (2003). *Sattipatthana: The direct path to realization*. Birmingham, England: Windhorse.
- Anderson, J. R., Bothell, D., Byrne, M. D., Douglas, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of mind. *Psychological Review*, 111, 1036–1060.
- Annels, S., Kho, K., & Bridge, P. (2016). Meditate don't meditate: How medical imaging evidence supports the role of meditation in the treatment of depression. *Radiography*, 22, e54–e58.
- Arch, J. J., Ayers, C. R., Baker, A., Almklov, E., Dean, D. J., & Craske, M. G. (2013). Randomized clinical trial of adapted mindfulness-based stress reduction versus group cognitive behavioral therapy for heterogeneous anxiety disorders. *Behaviour Research and Therapy*, 51, 185–196.
- Baer, R. A. (2011). Measuring mindfulness. *Contemporary Buddhism*, 12, 241–261. doi:10.1080/14639947.2011.564842
- Baer, R. A., Samuel, D. B., & Lykins, E. L. (2011). Differential item functioning on the Five Facet Mindfulness Questionnaire is minimal in demographically matched meditators and nonmeditators. *Assessment*, 18, 3–10.
- Baer, R. A., Smith, G. T., Hopkins, J., Kitemeyer, J., & Toney, L. (2006). Using self-report assessment methods to explore facets of mindfulness. *Assessment*, 13, 27–45.
- Baker, T. B., McFall, R. M., & Shoham, V. (2008). Current status and future prospects of clinical psychology toward a scientifically principled approach to mental and behavioral health care. *Psychological Science in the Public Interest*, 9, 67–103. doi:10.1111/j.1539-6053.2009.01036.x
- Baumeister, R. F., Vohs, K. D., & Funder, D. C. (2007). Psychology as the science of self-reports and finger movements: Whatever happened to actual behavior? *Perspectives on Psychological Science*, 2, 396–403. doi:10.1111/j.1745-6916.2007.00051.x
- Bent, S., Padula, A., & Avins, A. (2006). Better ways to question patients about adverse medical events. *Annals of Internal Medicine*, 144, 257–261.
- Bergomi, C., Tschacher, W., & Kupper, Z. (2013). The assessment of mindfulness with self-report measures: Existing scales and open issues. *Mindfulness*, 4, 191–202.
- Berkovich-Ohana, A., Dor-Ziderman, Y., Glicksohn, J., & Goldstein, A. (2013). Alterations in the sense of time, space, and body in the mindfulness-trained brain: A

- neurophenomenologically-guided MEG study. *Frontiers in Psychology*, 4, 912.
- Birnie, K., Garland, S. N., & Carlson, L. E. (2010). Psychological benefits for cancer patients and their partners participating in mindfulness-based stress reduction (MBSR). *Psycho-Oncology*, 9, 1004–1009.
- Bishop, S. R., Lau, M., Shapiro, S., Carlson, L., Anderson, N. D., Carmody, J., . . . Devins, G. (2004). Mindfulness: A proposed operational definition. *Clinical Psychology: Science and Practice*, 11, 230–241. doi:10.1093/clipsy.bph077
- Bodhi, B. (2011). What does mindfulness really mean? A canonical perspective. *Contemporary Buddhism*, 12, 19–39.
- Boorstein, S. (1996). Clinical aspects of meditation. In B. Scotton, A. Chinen, & J. Battista (Eds.), *Textbook of transpersonal psychiatry and psychology* (pp. 344–354). New York, NY: Basic Books.
- Boubela, R. N., Kalcher, K., Huf, W., Seidel, E.-M., Derntl, B., Pezawas, L., . . . Moser, E. (2015). fMRI measurements of amygdala activation are confounded by stimulus correlated signal fluctuation in nearby veins draining distance brain regions. *Scientific Reports*, 5, 10499. doi:10.1038/srep10499
- Braun, E. (2013). *The birth of insight: Meditation, modern Buddhism, and the Burmese Monk Ledi Sayadaw*. Chicago, IL: University of Chicago Press.
- Brefczynski-Lewis, J. A., Lutz, A., Schaefer, H. S., Levinson, D. B., & Davidson, R. J. (2007). Neural correlates of attentional expertise in long-term meditation practitioners. *Proceedings of the National Academy of Sciences USA*, 104, 11483–11488.
- Brewer, J. A., Worhunsky, P. D., Gray, J. R., Tang, Y.-Y., Weber, J., & Kober, H. (2011). Meditation experience is associated with differences in default mode network activity and connectivity. *Proceedings of the National Academy of Sciences USA*, 108, 20254–20259. doi:10.1073/pnas.1112029108
- Brewin, C. R. (2015). Re-experiencing traumatic events in PTSD: New avenues in research on intrusive memories and flashbacks. *European Journal of Psychotraumatology*, 6, 27180.
- Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications. *Psychological Review*, 117, 210–232.
- Briggs, J., & Killen, J. (2013). Perspectives on complementary and alternative medicine research. *Journal of the American Medical Association*, 310, 691–692.
- Britton, W. B. (2016). Scientific literacy as a foundational competency for teachers of mindfulness-based interventions. In D. McCown, D. K. Reibel, & M. S. Miccozzi (Eds.), *Resources for teaching mindfulness: A cross-cultural and international handbook* (pp. 93–119). New York, NY: Springer.
- Britton, W. B., Lepp, N. E., Niles, H. F., Rocha, T., Fisher, N., & Gold, J. (2014). A randomized controlled pilot trial of classroom-based mindfulness meditation compared to an active control condition in 6th grade children. *Journal of School Psychology*, 52, 263–278.
- Britton, W. B., Lindahl, J. R., Cahn, B. R., Davis, J. H., & Goldman, R. E. (2014). Awakening is not a metaphor: The effects of Buddhist meditation practices on basic wakefulness. *Annals of the New York Academy of Sciences*, 1307, 64–81. doi:10.1111/nyas.12279
- Brown, K. W., & Ryan, R. M. (2003). The benefits of being present: Mindfulness and its role in psychological well-being. *Journal of Personality and Social Psychology*, 84, 822–848. doi:10.1037/0022-3514.84.4.822
- Brown, K. W., Ryan, R. M., & Creswell, J. D. (2007). Addressing fundamental questions about mindfulness. *Psychological Inquiry*, 18, 211–237.
- Buddhaghosa, B. (1991). *The path of purification* (B. Nanamoli, Trans.). Onalaska, WA: Buddhist Publication Society.
- Bunning, S., & Blanke, O. (2005). The out-of-body experience: Precipitating factors and neural correlates. *Progress in Brain Research*, 150, 331–350.
- Button, K. S., Ioannidis, J. P., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S., & Munafò, M. R. (2013). Power failure: Why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience*, 14, 365–376.
- Carrington, P. (1977). *The misuse of meditation: Problems from overmeditation to freedom in meditation*. Garden City, NY: Anchor Books.
- Castellanos, F. X., Di Martino, A., Craddock, R. C., Mehta, A. D., & Milham, M. P. (2013). Clinical applications of the functional connectome. *NeuroImage*, 80, 527–540. doi:10.1016/j.neuroimage.2013.04.083
- Castillo, R. (1990). Depersonalization and meditation. *Psychiatry Research: Neuroimaging Section*, 53, 158–168.
- Cavanaugh, K., Strauss, C., Cicconi, F., Griffiths, N., Wyper, A., & Jones, F. (2013). A randomised controlled trial of a brief online mindfulness-based intervention. *Behaviour Research and Therapy*, 51, 573–578. doi:10.1016/j.brat.2013.06.003
- Chan, A. L. (2013, April 8). 20 reasons to love mindfulness (according to science). *Huffington Post*. Retrieved from http://www.huffingtonpost.com/2013/04/08/mindfulness-meditation-benefits-health_n_3016045.html
- Chan-Ob, T., & Boonyanaruthee, V. (1999). Meditation in association with psychosis. *Journal of the Medical Association of Thailand*, 82, 925–930.
- Chiesa, A., Calati, R., & Serretti, A. (2011). Does mindfulness training improve cognitive abilities? A systematic review of neuropsychological findings. *Clinical Psychology Review*, 31, 449–464. doi:10.1016/j.cpr.2010.11.003
- Christopher, M. S., Charoensuk, S., Gilbert, B. D., Neary, T. J., & Pearce, K. L. (2009). Mindfulness in Thailand and the United States: A case of apples versus oranges? *Journal of Clinical Psychology*, 65, 590–612. doi:10.1002/jclp.20580
- Cohen, A. S., Barlow, D. H., & Blanchard, E. B. (1985). Psychophysiology of relaxation-associated panic attacks. *Journal of Abnormal Psychology*, 94, 96–101.
- Coronado-Montoya, S., Levis, A. W., Kwakkenbos, L., Steele, R. J., Turner, E. H., & Thombs, B. D. (2016). Reporting of positive results in randomized controlled trials of mindfulness-based mental health interventions. *PLOS ONE*, 11, e0153220. doi:10.1371/journal.pone.0153220
- Cotman, C. W., Berchtold, N. C., & Christie, L.-A. (2007). Exercise builds brain health: Key roles of growth factor

- cascades and inflammation. *Trends in Neurosciences*, 30, 464–472. doi:10.1016/j.tins.2007.06.011
- Coyne, J. (2015a, March 30). Amazingly spun mindfulness trial in *British Journal of Psychiatry*: How to publish a null trial. *PLOS*. Retrieved from <http://blogs.plos.org/mindthebrain/2015/03/30/amazingly-spun-mindfulness-trial-in-british-journal-of-psychiatry-how-to-publish-a-null-trial/>
- Coyne, J. (2015b, May 20). Is mindfulness-based therapy ready for rollout to prevent relapse and recurrence in depression? *PLOS*. Retrieved from <http://blogs.plos.org/mindthebrain/2015/05/20/is-mindfulness-based-therapy-ready-for-rollout-to-prevent-relapse-and-recurrence-in-depression/>
- Coyne, J. (2016, November 16). Unintended consequences of universal mindfulness training for schoolchildren? *PLOS*. Retrieved from <http://blogs.plos.org/mindthebrain/2016/11/16/unintended-consequences-of-universal-mindfulness-training-for-schoolchildren/>
- Crane, R. S., & Kuyken, W. (2012). The implementation of mindfulness-based cognitive therapy: Learning from the UK health service experience. *Mindfulness*, 4, 246–254. doi:10.1007/s12671-012-0121-6
- Crawford, M. J., Thana, L., Farquharson, L., Palmer, L., Hancock, E., Bassett, P., . . . Parry, G. D. (2016). Patient experience of negative effects of psychological treatment: Results of a national surveydagger. *British Journal of Psychiatry*, 208, 260–265.
- Cumming, G. (2014). The new statistics: Why and how. *Psychological Science*, 25, 7–29.
- Dane, E. (2011). Paying attention to mindfulness and its effects on task performance in the workplace. *Journal of Management*, 37, 997–1018.
- Davidson, R. J. (2010). Empirical explorations of mindfulness: Conceptual and methodological conundrums. *Emotion*, 10, 8–11. doi:10.1037/a0018480
- Davidson, R. J., & Kaszniak, A. W. (2015). Conceptual and methodological issues in research on mindfulness and meditation. *American Psychologist*, 70, 581–592. doi:10.1037/a0039512
- Davidson, R. J., & Lutz, A. (2008). Buddha's brain: Neuroplasticity and meditation. *IEEE Signal Processing Magazine*, 25, 174–176.
- Didonna, F., & Gonzalez, Y. R. (2009). Mindfulness and feelings of emptiness. In F. Didonna (Ed.), *Clinical handbook of mindfulness* (pp. 125–151). New York, NY: Springer.
- Dimidjian, S., Beck, A., Felder, J. N., Boggs, J. M., Gallop, R., & Segal, Z. V. (2014). Web-based mindfulness-based cognitive therapy for reducing residual depressive symptoms: An open trial and quasi-experimental comparison to propensity score matched controls. *Behaviour Research and Therapy*, 63, 83–89. doi:10.1016/j.brat.2014.09.004
- Dimidjian, S., & Hollon, S. D. (2010). How would we know if psychotherapy were harmful? *American Psychologist*, 65, 21–33. doi:10.1037/a0017299
- Dimidjian, S., & Segal, S. V. (2015). Prospects for a clinical science of mindfulness-based intervention. *American Psychologist*, 70, 593–620.
- Disayavanish, C., & Disayavanish, P. (1984). Meditation-induced psychosis (in Thai). *Journal of the Psychiatric Association of Thailand*, 29, 1–12.
- Dobkin, P. L., Irving, J. A., & Amar, S. (2011). For whom may participation in a mindfulness-based stress reduction program be contraindicated? *Mindfulness*, 3, 44–50. doi:10.1007/s12671-011-0079-9
- Dor-Ziderman, Y., Berkovich-Ohana, A., Glicksohn, J., & Goldstein, A. (2013). Mindfulness-induced selflessness: A MEG neurophenomenological study. *Frontiers in Human Neuroscience*, 7, 582. doi:10.3389/fnhum.2013.00582
- Draganski, B., & May, A. (2008). Training-induced structural changes in the adult human brain. *Behavioural Brain Research*, 192, 137–142.
- Dreyfus, G. (2011). Is mindfulness present-centered and non-judgmental? A discussion of the cognitive dimensions of mindfulness. *Contemporary Buddhism*, 12, 41–54.
- Dunne, J. (2011). Toward an understanding of non-dual mindfulness. *Contemporary Buddhism*, 12, 71–88.
- Edinger, J., & Jacobsen, R. (1982). The incidence and significance of relaxation treatment side effects. *Behavior Therapist*, 5, 137–138.
- Eisendrath, S. J., Delucchi, K., Bitner, R., Fenimore, P., Smit, M., & McLane, M. (2008). Mindfulness-based cognitive therapy for treatment-resistant depression: A pilot study. *Psychotherapy and Psychosomatics*, 77, 319–320. doi:10.1159/000142525
- Embretson, S. (1983). Construct validity: Construct representation versus nomothetic span. *Psychological Bulletin*, 93, 179–197.
- Epstein, M., & Lieff, J. (1981). Psychiatric complications of meditation practice. *Journal of Transpersonal Psychology*, 13, 137–147.
- Fanelli, D. (2010). “Positive” results increase down the hierarchy of the sciences. *PLOS ONE*, 5(4), e10068. doi:10.1371/journal.pone.0010068
- Farb, N. (2014). From retreat center to clinic to boardroom? Perils and promises of the modern mindfulness movement. *Religions*, 5, 1062–1086.
- Farb, N., Segal, Z. V., Mayberg, H., Bean, J., McKeon, D., Fatima, Z., & Anderson, A. (2007). Attending to the present: Mindfulness meditation reveals distinct neural modes of self-reference. *Social Cognitive and Affective Neuroscience*, 2, 313–322.
- Farias, M., & Wikholm, C. (2015). *The Buddha Pill: Can meditation change you?* London, England: Watkins.
- Fenwick, P. (1983). Can we still recommend meditation? *British Medical Journal*, 287, 1401.
- Ferrarelli, F., Smith, R., Dentico, D., Riedner, B. A., Zenning, C., Benca, R., . . . Tononi, G. (2013). Experienced mindfulness meditators exhibit higher parietal-occipital EEG gamma activity during NREM sleep. *PLOS ONE*, 8, e73417. doi:10.1371/journal.pone.0073417
- Firestone, L. (2013, March). Benefits of mindfulness. *Psychology Today*. Retrieved from <http://www.psychologytoday.com/blog/compassion-matters/201303/benefits-mindfulness>
- Fox, K. C. R., Dixon, M. L., Nijeboer, S., Floman, J. L., Girn, M., Lifshitz, M., . . . Christoff, K. (2016). Functional

- neuroanatomy of meditation: A systematic review and meta-analysis of 78 functional neuroimaging investigations. *Neuroscience & Biobehavioral Reviews*, 65, 208–228.
- Fox, K. C. R., Nijeboer, S., Dixon, M. L., Floman, J. L., Ellamil, M., Rumak, S. P., . . . Christoff, K. (2014). Is meditation associated with altered brain structure? A systematic review and meta-analysis of morphometric neuroimaging in meditation practitioners. *Neuroscience & Biobehavioral Reviews*, 43, 48–73. doi:10.1016/j.neubiorev.2014.03.016
- Fox, K. C. R., Zakarauskas, P., Dixon, M., Ellamil, M., Thompson, E., & Christoff, K. (2012). Meditation experience predicts introspective accuracy. *PLOS ONE*, 7(9), e45370. doi:10.1371/journal.pone.0045370
- Freedman, D. H. (2010, November). Lies, damned lies, and medical science. *Atlantic*, 306, 76–84.
- Freeman, D., & Freeman, J. (2015, April). New study shows mindfulness therapy can be as effective as antidepressants. *Guardian*. Retrieved from <https://www.theguardian.com/science/blog/2015/apr/21/could-mindfulness-therapy-be-an-alternative-to-antidepressants>
- Frewen, P. A., Evans, E. M., Maraj, N., Dozois, D. J. A., & Partridge, K. (2007). Letting go: Mindfulness and negative automatic thinking. *Cognitive Therapy and Research*, 32, 758–774. doi:10.1007/s10608-007-9142-1
- Frewen, P. A., Lundberg, E., MacKinley, J., & Wrath, A. (2011). Assessment of response to mindfulness meditation: Meditation breath attention scores in association with subjective measures of state and trait mindfulness and difficulty letting go of depressive cognition. *Mindfulness*, 2, 254–269. doi:10.1007/s12671-011-0069-y
- Friston, K. (2012). Ten ironic rules for non-statistical reviewers. *NeuroImage*, 61, 1300–1310.
- Garland, E. L., Farb, N. A., Goldin, P. R., & Fredrickson, B. L. (2015). Mindfulness broadens awareness and builds eudaimonic meaning: A process model of mindful positive emotion regulation. *Psychological Inquiry*, 26, 293–314.
- Germer, C. K. (2005). Mindfulness: What is it? What does it matter? In C. K. Germer, R. D. Siegel, & P. R. Fulton (Eds.), *Mindfulness and psychotherapy* (pp. 3–28). London, England: Guilford.
- Geschwind, N., Peeters, F., Huibers, M., van Os, J., & Wichers, M. (2012). Efficacy of mindfulness-based cognitive therapy in relation to prior history of depression: Randomised controlled trial. *British Journal of Psychiatry*, 201, 320–325. doi:10.1192/bjp.bp.111.104851
- Gethin, R. (2011). On some definitions of mindfulness. *Contemporary Buddhism*, 12, 263–279.
- Gibbs, N. (Ed.). (2016, September). Mindfulness: The new science of health and happiness. *Time*. Retrieved from <https://shop.time.com/storefront/books/mindfulness-the-new-science-of-health-and-happiness/prodTDSHOP-MINDFULBZ.html>
- Giluk, T. L. (2009). Mindfulness, Big Five personality, and affect: A meta-analysis. *Personality and Individual Differences*, 47, 805–811.
- Goldberg, S. B., Wielgosz, J., Dahl, C., Shuyler, B., MacCoon, D. S., Rosenkranz, M., . . . Davidson, R. J. (2015). Does the Five Facet Mindfulness Questionnaire measure what we think it does? Construct validity evidence from an active controlled randomized clinical trial. *Psychological Assessment*, 28, 1009–1014.
- Goldin, P. R., & Gross, J. J. (2010). Effects of mindfulness-based stress reduction (MBSR) on emotion regulation in social anxiety disorder. *Emotion*, 10, 83–91.
- Goldin, P. R., Morrison, A., Jazaieri, H., Brozovich, F., Heimberg, R., & Gross, J. J. (2016). Group CBT versus MBSR for social anxiety disorder: A randomized controlled trial. *Journal of Consulting and Clinical Psychology*, 84, 427–437.
- Goleman, D. (1988). *The meditative mind: The varieties of meditative experience*. New York, NY: Tarcher.
- Goyal, M., Singh, S., Sibinga, E. M., Gould, N. F., Rowland-Seymour, A., Sharma, R., . . . Shihab, H. M. (2014). Meditation programs for psychological stress and well-being: A systematic review and meta-analysis. *JAMA Internal Medicine*, 174, 357–368.
- Grabovac, A., Lau, M., & Willett, B. (2011). Mechanisms of mindfulness: A Buddhist psychological model. *Mindfulness*, 2, 154–166. doi:10.1007/s12671-011-0054-5
- Green, C. D. (1992). Of immortal mythological beasts: Operationsim in psychology. *Theory & Psychology*, 2, 291–320.
- Greenberg, M. T., & Harris, A. R. (2012). Nurturing mindfulness in children and youth: Current state of research. *Child Development Perspectives*, 6, 161–166.
- Greene, D. J., Black, K. J., & Schlaggar, B. L. (2016). Considerations for MRI study design and implementation in pediatric and clinical populations. *Developmental Cognitive Neuroscience*, 18, 101–112.
- Grepmaier, L., Mitterlehner, F., Loew, T., Bachler, E., Rother, W., & Nickel, M. (2007). Promoting mindfulness in psychotherapists in training influences the treatment results of their patients: A randomized, double-blind, controlled study. *Psychotherapy and Psychosomatics*, 76, 332–338. doi:10.1159/000107560
- Grossman, P. (2011). Defining mindfulness by how poorly I think I pay attention during everyday awareness and other intractable problems for psychology's (re)invention of mindfulness: Comment on Brown et al. (2011). *Psychological Assessment*, 23, 1034–1040. doi:10.1037/a0022713
- Grossman, P., & Van Dam, N. T. (2011). Mindfulness, by any other name . . . : Trials and tribulations of sati in Western psychology and science. *Contemporary Buddhism*, 12, 219–239. doi:10.1080/14639947.2011.564841
- Gu, J., Strauss, C., Crane, C., Barnhofer, T., Karl, A., Cavanaugh, K., & Kuyken, W. (2016). Examining the factor structure of the 39-item and 15-item versions of the Five Facet Mindfulness Questionnaire before and after mindfulness-based cognitive therapy for people with recurrent depression. *Psychological Assessment*, 28, 791–802.
- Gunaratana, H. (2002). *Mindfulness in plain English*. Boston, MA: Wisdom.
- Gunderson, G. (2016, June 28). The science is in, and meditation may be the next big business opportunity. *Forbes*. Retrieved from <http://www.forbes.com/sites/garrett-gunderson/2016/06/28/the-science-is-in-and-meditation-may-be-the-next-big-business-opportunity/#2f729c7123c0>

- Hanley, A., Abell, N., Osborn, D., Roehrig, A., & Canto, A. (2016). Mind the gaps: Are conclusions about mindfulness entirely conclusive? *Journal of Counseling & Development*, 94, 103–113.
- Hanson, R. (2013). *Hardwiring happiness: The new brain science of contentment, calm, and confidence*. New York, NY: Harmony Books.
- Hasenkamp, W., & Barsalou, L. W. (2012). Effects of meditation experience on functional connectivity of distributed brain networks. *Frontiers in Human Neuroscience*, 6, 38. doi:10.3389/fnhum.2012.00038
- Hasenkamp, W., Wilson-Mendenhall, C. D., Duncan, E., & Barsalou, L. W. (2012). Mind wandering and attention during focused meditation: A fine-grained temporal analysis of fluctuating cognitive states. *NeuroImage*, 59, 750–760. doi:10.1016/j.neuroimage.2011.07.008
- Hayes, S. C. (2002). Buddhism and acceptance and commitment therapy. *Cognitive and Behavioral Practice*, 9, 58–66.
- Hayes, S. C., Strosahl, K. D., & Wilson, K. G. (1999). *Acceptance and commitment therapy: An experiential approach to behavior change*. New York, NY: Guilford.
- Heide, F., & Borkovec, T. (1983). Relaxation-induced anxiety: Paradoxical anxiety enhancement due to relaxation treatment. *Journal of Consulting and Clinical Psychology*, 51, 171–182.
- Hill, A. B. (1965). The environment and disease: Association or causation? *Proceedings of the Royal Society of Medicine*, 58, 295–300.
- Hofmann, S. G., & Asmundson, G. J. (2008). Acceptance and mindfulness-based therapy: New wave or old hat? *Clinical Psychology Review*, 28, 1–16.
- Hofmann, S. G., Sawyer, A. T., Witt, A. A., & Oh, D. (2010). The effect of mindfulness-based therapy on anxiety and depression: A meta-analytic review. *Journal of Consulting and Clinical Psychology*, 78, 169–183.
- Holmes, D. S., Solomon, S., Cappel, B. M., & Greenberg, J. L. (1983). Effects of transcendental meditation versus resting on physiological and subjective arousal. *Journal of Personality and Social Psychology*, 44, 1245–1252.
- Hölzel, B., Lazar, S., Gard, T., Schuman-Olivier, Z., Vago, D., & Ott, U. (2011). How does mindfulness meditation work? Proposing mechanisms of action from a conceptual and neural perspective. *Perspectives on Psychological Science*, 6, 537–559.
- Huffington, A. (2013, March 16). Mindfulness, meditation, wellness and their connection to corporate America's bottom line. *Huffington Post*. Retrieved from http://www.huffingtonpost.com/arianna-huffington/corporate-wellness_b_2903222.html
- Hupé, J. (2015). Statistical inferences under the Null hypothesis: Common mistakes and pitfalls in neuroimaging studies. *Frontiers in Neuroscience*, 9, 18.
- Hyde, K. L., Lerch, J., Norton, A., Forgeard, M., Winner, E., Evans, A. C., & Schlaug, G. (2009). Musical training shapes structural brain development. *Journal of Neuroscience*, 29, 3019–3025.
- Ioannidis, J. P. (2005). Why most published research findings are false. *PLOS Medicine*, 2(8), e124.
- Ioannidis, J. P. (2012). Why science is not necessarily self-correcting. *Perspectives on Psychological Science*, 7, 645–654.
- Jain, S., Shapiro, S. L., Swanick, S., Roesch, S. C., Mills, P. J., Bell, I., . . . Schwartz, G. E. R. (2007). A randomized controlled trial of mindfulness meditation versus relaxation training: Effects on distress, positive states of mind, rumination, and distraction. *Annals of Behavioral Medicine*, 33, 11–21.
- Jensen, C. G., Vangkilde, S., Frokjaer, V., & Hasselbalch, S. G. (2012). Mindfulness training affects attention—Or is it attentional effort? *Journal of Experimental Psychology: General*, 141, 106–123. doi:10.1037/a0024931
- Jha, A. P., Krompinger, J., & Baime, M. J. (2007). Mindfulness training modifies subsystems of attention. *Cognitive, Affective, & Behavioral Neuroscience*, 7, 109–119.
- Johnson, G. (2014a, January 21). New truths that only one can see. *New York Times*, p. D1.
- Johnson, G. (2014b, March 7). When studies are wrong: A coda. *New York Times*. Retrieved from <http://www.nytimes.com/2014/03/07/science/when-studies-are-wrong-a-coda.html>
- Jonsson, U., Alaie, I., Parling, T., & Arnberg, F. K. (2014). Reporting of harms in randomized controlled trials of psychological interventions for mental and behavioral disorders: A review of current practice. *Contemporary Clinical Trials*, 38, 1–8.
- Kabat-Zinn, J. (1990). *Full catastrophe living: The program of the Stress Reduction Clinic at the University of Massachusetts Medical Center*. New York, NY: Dell.
- Kabat-Zinn, J. (2011). Some reflections on the origins of MBSR, skillful means, and the trouble with maps. *Contemporary Buddhism*, 12, 281–306.
- Kabat-Zinn, J., & Davidson, R. J. (2011). *The mind's own physician: A scientific dialogue with the Dalai Lama on the healing power of meditation*. Oakland, CA: New Harbinger.
- Kerr, C. E., Josyula, K., & Littenberg, R. (2011). Developing an observing attitude: An analysis of meditation diaries in an MBSR clinical trial. *Clinical Psychology & Psychotherapy*, 18, 80–93. doi:10.1002/cpp.700
- Ketay, S., Hamilton, H. K., Haas, B. W., & Simeon, D. (2014). Face processing in depersonalization: An fMRI study of the unfamiliar self. *Psychiatry Research*, 222, 107–110. doi:10.1016/j.psychres.2014.02.003
- Kornfield, J. (1979). Intensive insight meditation: A phenomenological study. *Journal of Transpersonal Psychology*, 11, 41–58.
- Kozasa, E. H., Sato, J. R., Lacerda, S. S., Barreiros, M. A. M., Radvany, J., Russell, T. A., . . . Amaro, E., Jr. (2012). Meditation training increases brain efficiency in an attention task. *NeuroImage*, 59, 745–749. doi:10.1016/j.neuroimage.2011.06.088
- Kragel, P. A., & LaBar, K. S. (2014). Advancing emotion theory with multivariate pattern classification. *Emotion Review*, 6, 160–174.
- Kuijpers, H. J., van der Heijden, F. M. M. A., Tuinier, S., & Verhoeven, W. M. A. (2007). Meditation-induced psychosis. *Psychopathology*, 40, 461–464. doi:10.1159/000108125
- Kutz, I., Leserman, J., Dorrington, C., Morrison, C. H., Borysenko, J. Z., & Benson, H. (1985). Meditation

- as an adjunct to psychotherapy. An outcome study. *Psychotherapy and Psychosomatics*, 43, 209–218.
- Kuyken, W., Crane, R., & Dalgleish, T. (2012). Does mindfulness based cognitive therapy prevent relapse of depression? *British Medical Journal*, 345, e7194. doi:10.1136/bmj.e7194
- Kuyken, W., Crane, W., & Williams, J. M. (2012). *Mindfulness-based cognitive therapy (MBCT) implementation resources*. Oxford, England: Oxford University, University of Exeter, Bangor University.
- Kuyken, W., Hayes, R., Barrett, B., Byng, R., Dalgleish, T., Kessler, D., . . . Byford, S. (2015). Effectiveness and cost-effectiveness of mindfulness-based cognitive therapy compared with maintenance antidepressant treatment in the prevention of depressive relapse or recurrence (PREVENT): A randomised controlled trial. *Lancet*, 386, 63–73.
- Kuyken, W., Warren, F. C., Taylor, R. S., Whalley, B., Crane, C., Bondolfi, G., . . . Dalgleish, T. (2016). Efficacy of mindfulness-based cognitive therapy in prevention of depressive relapse: An individual patient data meta-analysis from randomized trials. *JAMA Psychiatry*, 73, 565–574.
- Langer, E. J. (1989). *Mindfulness*. New York, NY: Perseus Books.
- Lao, S. A., Kissane, D., & Meadows, G. (2016). Cognitive effects of MBSR/MBCT: A systematic review of neuropsychological outcomes. *Consciousness and Cognition*, 45, 109–123.
- Lazar, S. W., Bush, G., Gollub, R. L., Fricchione, G. L., Khalsa, G., & Benson, H. (2000). Functional brain mapping of the relaxation response and meditation. *NeuroReport*, 11, 1581.
- Lehrer, J. (2010, December 13). The truth wears off. *New Yorker*, p. 52.
- Leigh, J., Bowen, S., & Marlatt, G. A. (2005). Spirituality, mindfulness, and substance abuse. *Addictive Behaviors*, 30, 1335–1341.
- Levinson, D. B., Stoll, E. L., Kindy, S. D., Merry, H. L., & Davidson, R. J. (2014). A mind you can count on: Validating breath counting as a behavioral measure of mindfulness. *Frontiers in Psychology*, 5, 1202. doi:10.3389/fpsyg.2014.01202
- Lilienfeld, S. O., Lynn, S. J., & Lohr, J. M. (2003). *Science and pseudoscience in clinical psychology*. New York, NY: Guilford.
- Lim, D., Condon, P., & DeSteno, D. (2015). Mindfulness and compassion: An examination of mechanism and scalability. *PLOS ONE*, 10(2), e0118221.
- Lindahl, J. R., Fisher, N. E., Cooper, D. J., Rosen, R. K., & Britton, W. B. (2017). The varieties of contemplative experience: A mixed-methods study of meditation-related challenges in Western Buddhists. *PLOS ONE*, 12(5), e0176239. https://doi.org/10.1371/journal.pone.0176239
- Lindahl, J. R., Kaplan, C., Winget, E., & Britton, W. B. (2014). A phenomenology of meditation-induced light experiences: Traditional Buddhist and neurobiological perspectives. *Frontiers in Psychology*, 4, 973.
- Linden, M. (2013). How to define, find and classify side effects in psychotherapy: From unwanted events to adverse treatment reactions. *Clinical Psychology & Psychotherapy*, 20, 286–296. doi:10.1002/cpp.1765
- Linden, W. (1990). *Autogenic training: A clinical guide*. New York, NY: Guilford.
- Linehan, M. (1993). *Cognitive behavioral treatment of borderline personality disorder*. New York, NY: Guilford.
- Lomas, T., Cartwright, T., Edginton, T., & Ridge, D. (2015). A qualitative analysis of experiential challenges associated with meditation practice. *Mindfulness*, 6, 848–860. doi:10.1007/s12671-014-0329-8
- Luders, E., Kurth, F., Toga, A. W., Narr, K. L., & Gaser, C. (2013). Meditation effects within the hippocampal complex revealed by voxel-based morphometry and cytoarchitectonic probabilistic mapping. *Frontiers in Psychology*, 4, 398.
- Luders, E., Thompson, P. M., Kurth, F., Hong, J., Phillips, O. R., Wang, Y., . . . Toga, A. W. (2012). Global and regional alterations of hippocampal anatomy in long-term meditation practitioners. *Human Brain Mapping*, 34, 3369–3375.
- Lustyk, M., Chawla, N., Nolan, R., & Marlatt, G. (2009). Mindfulness meditation in research: A discussion of safety issues and participant screening procedures. *Advances in Mind-Body Medicine*, 24, 20–30.
- Lutz, A., Dunne, J. D., & Davidson, R. J. (2007). Meditation and the neuroscience of consciousness: An introduction. In P. D. Zelazo, M. Moscovitch, & E. Thompson (Eds.), *The Cambridge handbook of consciousness* (pp. 499–551). New York, NY: Cambridge University Press.
- Lutz, A., Greischar, L. L., Perlman, D. M., & Davidson, R. J. (2009). BOLD signal in insula is differentially related to cardiac function during compassion meditation in experts vs. novices. *NeuroImage*, 47, 1038–1046.
- Lutz, A., Jha, A. P., Dunne, J. D., & Saron, C. D. (2015). Investigating the phenomenological matrix of mindfulness-related practices from a neurocognitive perspective. *American Psychologist*, 70, 632–658.
- Lutz, A., Lachaux, J. P., Martinerie, J., & Varela, F. J. (2002). Guiding the study of brain dynamics by using first-person data: Synchrony patterns correlate with ongoing conscious states during a simple visual task. *Proceedings of the National Academy of Sciences USA*, 99, 1586–1591.
- Lutz, A., Slagter, H. A., Dunne, J. D., & Davidson, R. J. (2008). Attention regulation and monitoring in meditation. *Trends in Cognitive Sciences*, 12, 163–169.
- Lutz, A., & Thompson, E. (2003). Neurophenomenology integrating subjective experience and brain dynamics in the neuroscience of consciousness. *Journal of Consciousness Studies*, 10, 31–52.
- MacCoon, D. G., Imel, Z. E., Rosenkranz, M. A., Sheftel, J. G., Wang, H. Y., Sullivan, J. C., . . . Lutz, A. (2012). The validation of an active control intervention for Mindfulness Based Stress Reduction (MBSR). *Behaviour Research and Therapy*, 50, 3–12.
- Mackey, A. P., Miller Singley, A. T., & Bunge, S. A. (2013). Intensive reasoning training alters patterns of brain connectivity at rest. *Journal of Neuroscience*, 33, 4796–4803.
- Malinowski, P. (2013). Neural mechanisms of attentional control in mindfulness meditation. *Frontiers in Human Neuroscience*, 7, 8. doi:10.3389/fnhn.2013.00008
- Manicavasagar, V., Parker, G., & Perich, T. (2011). Mindfulness-based cognitive therapy vs cognitive behaviour therapy

- as a treatment for non-melancholic depression. *Journal of Affective Disorders*, 130, 138–144.
- Manocha, R. (2000). Why meditation? *Australian Family Physician*, 29, 1135–1138.
- Manuel, J. A., Somohano, V. C., & Bowen, S. (2017). Mindfulness practice and its relationship to the Five-Facet Mindfulness Questionnaire. *Mindfulness*, 8, 361–367. doi:10.1007/s12671-016-0605-x
- Martin, D. J., Garske, J. P., & Davis, M. K. (2000). Relation of the therapeutic alliance with outcome and other variables: A meta-analytic review. *Journal of Consulting and Clinical Psychology*, 68, 438–450.
- Mascaro, J. S., Rilling, J. K., Negi, L. T., & Raison, C. L. (2013). Pre-existing brain function predicts subsequent practice of mindfulness and compassion meditation. *NeuroImage*, 69, 35–42.
- Matzke, D., Nieuwenhuis, S., van Rijn, H., Slagter, H. A., van der Molen, M. W., & Wagenmakers, E.-J. (2015). The effect of horizontal eye movements on free recall: A preregistered adversarial collaboration. *Journal of Experimental Psychology. General*, 144, e1–15. doi:10.1037/xge0000038
- McMahan, D. L. (2008). *The making of Buddhist modernism*. Oxford, England: Oxford University Press.
- Meyer, D. E. (2009, April). *Multi-tasking, meditation, and contemplative practice*. Presentation at Mind and Life XVIII: Attention, Memory and the Mind, Dharamsala, India.
- Meyer, D. E., & Kieras, D. E. (1999). Precis to a practical unified theory of cognition and action: Some lessons from EPIC computational models of human multiple-task performance. In D. Gopher & A. Koriati (Eds.), *Attention and Performance XVII. Cognitive regulation of performance: Interaction of theory and application* (pp. 17–88). Cambridge, MA: MIT Press.
- Miguel, E., Camerer, C., Casey, K., Cohen, J., Esterling, K. M., Gerber, A., . . . Van der Laan, M. (2014). Promoting transparency in social science research. *Science*, 343, 30–31. doi:10.1126/science.1245317
- Miller, J. (1993). The unveiling of traumatic memories and emotions through mindfulness and concentration meditation: Clinical implications and three case reports. *Journal of Transpersonal Psychology*, 25, 169–180.
- Moher, D., Schulz, K. F., & Altman, D. G. (2001). The CONSORT statement: Revised recommendations for improving the quality of reports of parallel-group randomized trials. *Lancet*, 357, 1191–1194.
- Mohr, D. C. (1995). Negative outcome in psychotherapy: A critical review. *Clinical Psychology: Science and Practice*, 2, 1–27.
- Moos, R. H. (2005). Iatrogenic effects of psychosocial interventions for substance use disorders: Prevalence, predictors, prevention. *Addiction*, 100, 595–604. doi:10.1111/j.1360-0443.2005.01073.x
- Moos, R. H. (2012). Iatrogenic effects of psychosocial interventions: Treatment, life context, and personal risk factors. *Substance Use & Misuse*, 47, 1592–1598. doi:10.3109/10826084.2012.705710
- Mrazek, M. D., Smallwood, J., & Schooler, J. W. (2012). Mindfulness and mind-wandering: Finding convergence through opposing constructs. *Emotion*, 12, 442–448. doi:10.1037/a0026678
- Munafò, M. R., Stothart, G., & Flint, J. (2009). Bias in genetic association studies and impact factor. *Molecular Psychiatry*, 14, 119–120. doi:10.1038/mp.2008.77
- Münste, T. F., Altenmüller, E., & Jäncke, L. (2002). The musician's brain as a model of neuroplasticity. *Nature Reviews Neuroscience*, 3, 473–478.
- Nakaya, M., & Ohmori, K. (2010). Psychosis induced by spiritual practice and resolution of pre-morbid inner conflicts. *German Journal of Psychiatry*, 13, 161–163.
- National Center for Complementary and Integrative Health. (2016a). *NCCIH Clinical Research Toolbox: Data safety monitoring*. National Center for Complementary and Integrative Health, National Institutes of Health. Retrieved from <https://nccih.nih.gov/grants/toolbox#DSM>
- National Center for Complementary and Integrative Health. (2016b). *What the science says about safety and side effects of meditation*. National Center for Complementary and Integrative Health, National Institutes of Health. Retrieved from <https://nccih.nih.gov/health/meditation/overview.htm#hed5>
- National Institutes of Health. (2016). *Adverse event and serious adverse event guidelines OHRP guidance on reviewing and reporting unanticipated problems involving risks to subjects or others and adverse events, OHRP guidance*. Bethesda, MD: National Institutes of Health, Office for Human Research Protections, U.S. Department of health and Human Services.
- National Institutes on Aging. (2011). *NIA adverse event and serious adverse event guidelines*. Bethesda, MD: National Institutes on Aging, National Institutes of Health.
- Neisser, U., Boodoo, G., Bouchard, T. J., Boykin, A. W., Brody, N., Ceci, S. J., . . . Urbina, S. (1996). Intelligence: Knowns and unknowns. *American Psychologist*, 51, 77–101.
- Nichols, T. E., Das, S., Eickhoff, S. B., Evans, A. C., Glatard, T., Hanke, M., . . . Yeo, B. T. T. (2017). Best practices in data analysis and sharing in neuroimaging using MRI. *Nature Neuroscience*, 20, 299–303.
- Nyham, B. (2014, September 18). To get more out of science, show the rejected research. *New York Times*. Retrieved from <http://www.nytimes.com/2014/09/19/upshot/to-get-more-out-of-science-show-the-rejected-research.html>
- Office for Human Research Protections. (2007). *Guidance on reviewing and reporting unanticipated problems involving risks to subjects or others and adverse events*. Washington, DC: U.S. Department of Health and Human Services, Office for Human Research Protections.
- Onken, L. S., Carroll, K. M., Shoham, V., Cuthbert, B. N., & Riddle, M. (2014). Reenvisioning clinical science: Unifying the discipline to improve public health. *Clinical Psychological Science*, 2, 22–34.
- Open Science Collaboration. (2012). An open, large-scale, collaborative effort to estimate the reproducibility of psychological science. *Perspectives on Psychological Science*, 7, 657–660.
- Ospina, M. B., Bond, K., Karkhanavich, M., Tjosvold, L., Vandermeer, B., Liang, Y., . . . Lassen, T. P. (2007).

- Meditation practices for health: State of the research* (Evidence Report/Technology Assessment). Rockville, MD: National Center for Complementary and Alternative Medicine. Retrieved from <https://www.ncbi.nlm.nih.gov/books/NBK38360/>
- Papies, E. K., Barsalou, L. W., & Custers, R. (2012). Mindful attention prevents mindless impulses. *Social Psychological & Personality Science*, 3, 291–299. doi:10.1177/1948550611419031
- Papies, E. K., Pronk, T. M., Keesman, M., & Barsalou, L. W. (2015). The benefits of simply observing: Mindful attention modulates the link between motivation and behavior. *Journal of Personality and Social Psychology*, 108, 148–170. doi:10.1037/a0038032
- Pashler, H., & Wagenmakers, E. (2012). Editors' introduction to the special section on replicability in psychological science: A crisis of confidence? *Perspectives on Psychological Science*, 7, 528–530.
- Penedo, F. J., & Dahn, J. R. (2005). Exercise and well-being: A review of mental and physical health benefits associated with physical activity. *Current Opinion in Psychiatry*, 18, 189–193.
- Plaud, J. J. (2001). Paradigms, promises, and the potential of clinical psychology. *Journal of Clinical Psychology*, 57, 1089–1102.
- Power, J. D., Cohen, A. L., Nelson, S. M., Wig, G. S., Barnes, K. A., Church, J. A., . . . Petersen, S. E. (2011). Functional network organization of the human brain. *Neuron*, 72, 665–678. doi:10.1016/j.neuron.2011.09.006
- Reuter, M., Tisdall, M. D., Qureshi, A., Buckner, R. L., van der Kouwe, A. J., & Fischl, B. (2015). Head motion during MRI acquisition reduces gray matter volume and thickness estimates. *NeuroImage*, 107, 107–115.
- Rhodes, E. (2015, September). Mindfulness on trial. *Psychologist*, 28(9). Retrieved from <https://thepsychologist.bps.org.uk/mindfulness-trial>
- Ring, C., Brener, J., Knapp, K., & Mailloux, J. (2015). Effects of heartbeat feedback on beliefs about heart rate and heartbeat counting: A cautionary tale about interoceptive awareness. *Biological Psychology*, 104, 193–198. doi:10.1016/j.biopsycho.2014.12.010
- Robins, C. J. (2002). Zen principles and mindfulness practice in dialectical behavior therapy. *Cognitive and Behavioral Practice*, 9, 50–57.
- Rosnow, R. L. (2002). The nature and role of demand characteristics in scientific inquiry. *Prevention & Treatment*, 5, 37. doi:10.1037/1522-3736.5.1.537c
- Rosnow, R. L., & Rosenthal, R. (1989). Statistical procedures and the justification of knowledge in psychological science. *American Psychologist*, 44, 1276–1284.
- Sahdra, B. K., MacLean, K. A., Ferrer, E., Shaver, P. R., Rosenberg, E. L., Jacobs, T. L., . . . Saron, C. D. (2011). Enhanced response inhibition during intensive meditation training predicts improvements in self-reported adaptive socioemotional functioning. *Emotion*, 11, 299–312.
- Santorelli, S. (2014). *Mindfulness-based stress reduction (MBSR): Standards of practice*. Worcester, MA: Center for Mindfulness in Medicine, Health Care, and Society, University of Massachusetts Medical School.
- Sauer, S., Walach, H., Schmidt, S., Hinterberger, T., Lynch, S., Bussing, A., & Kohls, N. (2013). Assessment of mindfulness: Review on state of the art. *Mindfulness*, 4, 3–17.
- Sayadaw, M. (1965). *The progress of insight: A modern Pali Treatise on Buddhist Satipatthana meditation* (Nyanaponika Thera, Trans.). Kandy, Sri Lanka: Buddhist Publication Society.
- Schaufenbuel, K. (2015, December 28). Why Google, Target, and General Mills are investing in mindfulness. *Harvard Business Review*. Retrieved from <https://hbr.org/2015/12/why-google-target-and-general-mills-are-investing-in-mindfulness>
- Sedlmeier, P., Eberth, J., Schwarz, M., Zimmermann, D., Haairig, F., Jaeger, S., & Kunze, S. (2012). The psychological effects of meditation: A meta-analysis. *Psychological Bulletin*, 138, 1139–1171. doi:10.1037/a0028168
- Segal, Z. V., Bieling, P., Young, T., MacQueen, G., Cooke, R., . . . Levitan, R. D. (2010). Antidepressant monotherapy vs sequential pharmacotherapy and mindfulness-based cognitive therapy, or placebo, for relapse prophylaxis in recurrent depression. *Archives of General Psychiatry*, 67, 1256–1264.
- Segal, Z. V., Williams, J. M. G., & Teasdale, J. D. (2002). *Mindfulness-based cognitive therapy for depression: A new approach to preventing relapse*. New York, NY: Guilford.
- Segal, Z. V., Williams, J. M. G., & Teasdale, J. D. (2012). *Mindfulness-based cognitive therapy for depression* (2nd ed.). New York, NY: Guilford.
- Sethi, S. (2003). Relationship of meditation and psychosis: Case studies. *Australian & New Zealand Journal of Psychiatry*, 37, 382.
- Shan, H. (2000). Culture-bound psychiatric disorders associated with qigong practice in China. *Hong Kong Journal of Psychiatry*, 10, 12–14.
- Shapiro, D. H., Jr. (1982). Overview: Clinical and physiological comparison of meditation with other self-control strategies. *American Journal of Psychiatry*, 139, 267–274.
- Shapiro, D. H., Jr. (1992). Adverse effects of meditation: A preliminary investigation of long-term meditators. *International Journal of Psychosomatics*, 39, 62–67.
- Shapiro, S. L., Carlson, L. E., Astin, J. A., & Freedman, B. (2006). Mechanisms of mindfulness. *Journal of Clinical Psychology*, 62, 373–386.
- Shonin, E., Van Gordon, W., & Griffiths, M. D. (2013). Mindfulness-based interventions: Towards mindful clinical integration. *Frontiers in Psychology*, 4, 194.
- Shonin, E., Van Gordon, W., & Griffiths, M. D. (2014a). Are there risks associated with using mindfulness in the treatment of psychopathology? *Clinical Practice*, 11, 389–392.
- Shonin, E., Van Gordon, W., & Griffiths, M. D. (2014b). Cognitive behavioral therapy (CBT) and meditation awareness training (MAT) for the treatment of co-occurring schizophrenia with pathological gambling: A case study. *International Journal of Mental Health and Addiction*, 12, 181–196.
- Shonin, E., Van Gordon, W., & Griffiths, M. D. (2014c). Do mindfulness-based therapies have a role in the treatment of psychosis? *Australian & New Zealand Journal of Psychiatry*, 48, 124–127. doi:10.1177/0004867413512688

- Shwed, U., & Bearman, P. S. (2010). The temporal structure of scientific consensus formation. *American Sociological Review*, 75, 817–840.
- Sierra, M., Senior, C., Dalton, J., McDonough, M., Bond, A., Phillips, M. L., . . . David, A. S. (2002). Autonomic response in depersonalization disorder. *Archives of General Psychiatry*, 59, 833–838.
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22, 1359–1366. doi:10.1177/0956797611417632
- Singh, N. N., Lancioni, G. E., Winton, A. S. W., Wahler, R. G., Singh, J., & Sage, M. (2004). Mindful caregiving increases happiness among individuals with profound multiple disabilities. *Research in Developmental Disabilities*, 25, 207–218. doi:10.1016/j.ridd.2003.05.001
- Slagter, H. A., Lutz, A., Greischar, L. L., Francis, A. D., Nieuwenhuis, S., Davis, J. M., & Davidson, R. J. (2007). Mental training affects distribution of limited brain resources. *PLOS Biology*, 5(6), e138.
- Sperduti, M., Martinelli, P., & Piolino, P. (2012). A neurocognitive model of meditation based on activation likelihood estimation (ALE) meta-analysis. *Consciousness and Cognition*, 21, 269–276.
- Strauss, C., Cavanagh, K., Oliver, A., & Pettman, D. (2014). Mindfulness-based interventions for people diagnosed with a current episode of an anxiety or depressive disorder. A meta-analysis of randomised controlled trials. *PLOS ONE*, 9(4), e96110. doi:10.1371/journal.pone.0096110
- Strauss, M. E., & Smith, G. T. (2009). Construct validity: Advances in theory and methodology. *Annual Review of Clinical Psychology*, 5, 1–25.
- Strawn, J. R., Cotton, S., Luberto, C. M., Patino, L. R., Stahl, L. A., Weber, W. A., . . . DelBello, M. P. (2016). Neural function before and after mindfulness-based cognitive therapy in anxious adolescents at risk for developing bipolar disorder. *Journal of Child and Adolescent Psychopharmacology*, 26, 372–379. doi:10.1089/cap.2015.0054
- Sze, J. A., Gyurak, A., Yuan, J. W., & Levenson, R. W. (2010). Coherence between emotional experience and physiology: Does body awareness training have an impact? *Emotion*, 10, 803–814.
- Tang, Y.-Y., Hölzel, B. K., & Posner, M. I. (2015). The neuroscience of mindfulness meditation. *Nature Reviews Neuroscience*, 16, 213–225. doi:10.1038/nrn3916
- Tang, Y.-Y., Ma, Y., Wang, J., Fan, Y., Feng, S., Lu, Q., . . . Posner, M. I. (2007). Short-term meditation training improves attention and self-regulation. *Proceedings of the National Academy of Sciences USA*, 104, 17152–17156.
- Tolin, D. F. (2010). Is cognitive-behavioral therapy more effective than other therapies? A meta-analytic review. *Clinical Psychology Review*, 30, 710–720.
- Tomasino, B., Fregona, S., Skrap, M., & Fabbro, F. (2013). Meditation-related activations are modulated by the practice needed to obtain it and by the expertise: An ALE meta-analysis study. *Frontiers in Human Neuroscience*, 6, 346. doi:10.3389/fnhum.2012.00346
- Turner, L., Singh, K., Garrity, C., Tsertsvadze, A., Manheimer, E., Wieland, L., . . . Moher, D. (2011). An evaluation of the completeness of safety reporting in reports of complementary and alternative medicine trials. *BMC Complementary and Alternative Medicine*, 11, 67.
- Turner, R. (2002). How much cortex can a vein drain? Downstream dilution of activation-related cerebral blood oxygenation changes. *NeuroImage*, 16, 1062–1067.
- Vago, D. R., & Silbersweig, D. A. (2012). Self-awareness, self-regulation, and Self-transcendence (S-ART): A framework for understanding the neurobiological mechanisms of mindfulness. *Frontiers in Human Neuroscience*, 6, 00296. doi:10.3389/fnhum.2012.00296
- van Aalderen, J. R., Donders, A. R. T., Giommi, F., Spinhoven, P., Barendregt, H. P., & Speckens, A. E. M. (2012). The efficacy of mindfulness-based cognitive therapy in recurrent depressed patients with and without a current depressive episode: A randomized controlled trial. *Psychological Medicine*, 42, 989–1001. doi:10.1017/S0033291711002054
- Van Dam, N. T., Earleywine, M., & Borders, A. (2010). Measuring mindfulness? An item response theory analysis of the Mindful Attention Awareness Scale. *Personality and Individual Differences*, 49, 805–810. doi:10.1016/j.paid.2010.07.020
- Van Dam, N. T., Earleywine, M., & Danoff-Burg, S. (2009). Differential item function across meditators and non-meditators on the Five Facet Mindfulness Questionnaire. *Personality and Individual Differences*, 47, 516–521. doi:10.1016/j.paid.2009.05.005
- Van den Hurk, P. A. M., Giommi, F., Gielen, S. C., Speckens, A. E. M., & Barendregt, H. P. (2010). Greater efficiency in attentional processing related to mindfulness meditation. *Quarterly Journal of Experimental Psychology*, 63, 1168–1180.
- VanderKooi, L. (1997). Buddhist teachers' experience with extreme mental states in Western meditators. *Journal of Transpersonal Psychology*, 29, 31–46.
- Van Dijk, K. R. A., Sabuncu, M. R., & Buckner, R. L. (2012). The influence of head motion on intrinsic functional connectivity MRI. *NeuroImage*, 59, 431–438. doi:10.1016/j.neuroimage.2011.07.044
- Van Nuys, D. (1973). Meditation, attention, and hypnotic susceptibility: A correlation study. *International Journal of Clinical and Experimental Hypnosis*, 21, 59–69.
- van Vugt, M. K., & Slagter, H. A. (2014). Control over experience? Magnitude of the attentional blink depends on meditative state. *Consciousness and Cognition*, 23, 32–39. doi:10.1016/j.concog.2013.11.001
- van Vugt, M. K., Taatgen, N. A., Bastian, M., & Sackur, J. (2015, April). *Modeling mind-wandering: A tool to better understand distraction*. Paper presented at the International Conference in Cognitive Modeling, Groningen, Netherlands.
- Vaughan, B., Goldstein, M. H., Alikakos, M., Cohen, L. J., & Serby, M. J. (2014). Frequency of reporting of adverse events in randomized controlled trials of psychotherapy vs. psychopharmacotherapy. *Comprehensive Psychiatry*, 55, 849–855. doi:10.1016/j.comppsy.2014.01.001

- Wallace, B. (2011). *Stilling the mind: Shamatha teachings from Dudjom Lingpa's Vajra essence*. Boston, MA: Wisdom Publications.
- Wallace, R. K. (1970). Physiological effects of transcendental meditation. *Science*, 167, 1751–1754.
- Wallace, R. K., Benson, H., & Wilson, A. F. (1971). A wakeful hypometabolic physiologic state. *American Journal of Physiology*, 221, 795–799.
- Walsh, R. (2015). What is wisdom? Cross-cultural and cross-disciplinary syntheses. *Review of General Psychology*, 19, 278–293.
- Walsh, R., & Roche, L. (1979). Precipitation of acute psychotic episodes by intensive meditation in individuals with a history of schizophrenia. *American Journal of Psychiatry*, 136, 1085–1086.
- Walton, A. (2014, January 7). For depression treatment, meditation might rival medication. *Forbes*, pp. 1–3.
- Whitmarsh, S., Barendregt, H., Schoffelen, J.-M., & Jensen, O. (2014). Metacognitive awareness of covert somatosensory attention corresponds to contralateral alpha power. *NeuroImage*, 85, 803–809. doi:10.1016/j.neuroimage.2013.07.031
- Williams, J. M., Crane, C., Barnhofer, T., Brennan, K., Duggan, D. S., Fennell, M. J., . . . Russell, I. T. (2014). Mindfulness-based cognitive therapy for preventing relapse in recurrent depression: A randomized dismantling trial. *Journal of Consulting and Clinical Psychology*, 82, 275–286. doi:10.1037/a0035036
- World Health Organization. (2016). *The use of the WHO-UMC system for standardized case causality assessment*. World Health Organization, Uppsala Monitoring Centre. Retrieved from who-umc.org
- Yarkoni, T., Poldrack, R. A., Van Essen, D. C., & Wager, T. D. (2010). Cognitive neuroscience 2.0: Building a cumulative science of human brain function. *Trends in Cognitive Sciences*, 14, 489–496. doi:10.1016/j.tics.2010.08.004
- Yorston, G. (2001). Mania precipitated by meditation: A case report and literature review. *Mental Health, Religion & Culture*, 4, 209–214.
- Zanesco, A. P., King, B. G., MacLean, K. A., & Saron, C. D. (2013). Executive control and felt concentrative engagement following intensive meditation training. *Frontiers in Human Neuroscience*, 7, 00566. doi:10.3389/fnhum.2013.00566
- Zeidan, F., Emerson, N. M., Farris, S. R., Ray, J. N., Jung, Y., McHaffie, J. G., & Coghill, R. C. (2015). Mindfulness meditation-based pain relief employs different neural mechanisms than placebo and sham mindfulness meditation-induced analgesia. *Journal of Neuroscience*, 35, 15307–15325. doi:10.1523/JNEUROSCI.2542-15.2015
- Zeidan, F., Martucci, K. T., Kraft, R. A., Gordon, N. S., McHaffie, J. G., & Coghill, R. C. (2011). Brain mechanisms supporting the modulation of pain by mindfulness meditation. *Journal of Neuroscience*, 31, 5540–5548.